A Discrete Choice Model for Financial Product Demand Estimation and An Application on Quantifying the Willingness to Pay for Fintech*

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Abstract

One underlying assumption in the well-known BLP model (Berry et al., 1995) is that all consumers in a market face the same price for the same product. However, in financial markets, different consumers with different demographic characteristics face different prices for the same product (individual-risk-based pricing). The effect of income on price sensitivities will be overestimated if researchers simply apply BLP. We propose a two-step estimation for financial-product demand with bias correction for selectivity that individual-specific prices of only products chosen by individuals are observed. We apply the method to mortgage markets. The estimate of the income effect on price sensitivities generated by the traditional BLP is 11.20% greater than that generated by our method. We further quantify borrowers' willingness to pay for fintech and the value of face-to-face interaction for first-time borrowers.

Keywords: Discrete Choice Model, Demand Estimation, Differentiated Product, Financial Product, Fintech, Mortgage, Bank, Willingness to Pay

JEL Codes: C01, D14, G21, G23, G51, L10, L80, M31

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1. Introduction

Demand estimation for differentiated products is the fundamental component of empirical analyses in modern industrial organization (IO). It provides bases for examining many important questions, including market power, mergers, entry, consumers' willingness to pay, pricing, regulation, and deregulation. The methodology developed in the seminal papers Berry (1994) and Berry et al. (1995) (henceforth, BLP) has been used in analyzing market outcomes and policy issues in many industries (see Berry and Haile, 2016 for a review of empirical studies applying the BLP framework).

However, research that applied the BLP framework to analyses of financial-product industries is limited, although many financial-product industries have differentiated products. For example, mortgages are differentiated in several dimensions, including lender, maturity, fixed-rate vs. adjustable-rate, with or without prepayment penalty, upfront points, loan-to-value ratio (LTV), etc. Other examples include student loans, automobile loans, personal loans, home equity loans, credit cards, and reverse mortgages. Financial-product industries constitute an essential sector in our economy with vast sales volumes and many important policy concerns. For example, mortgage originations in the U.S. during 2022 amounted to approximately 4 trillion dollars, compared to 0.86 trillion dollars for the U.S. motor vehicle and parts manufacturing gross output in the same year.

One underlying assumption in the BLP model (even the micro-BLP model) is that all consumers in a market face the same price for the same product. Accordingly, average transaction prices, manufacturer's suggested retail prices (MSRP), or listed prices are used as product prices faced by all consumers. This treatment is natural for nonfinancial-product industries (e.g., the automobile and grocery industries). However, in financial-product industries, different consumers usually face different prices for the same financial product, and the price a consumer faces is related to the consumer's demographic characteristics (individual-risk-based pricing).[1] Meanwhile, the consumer's price elasticity of demand is also related to its demographic characteristics. For example, mortgage borrowers with lower incomes tend to be charged higher prices (interest rates)

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[1] Empirical evidence on lenders' individual-risk-based pricing in the mortgage and consumer loan markets has been provided by Edelberg (2006) and Magri and Pico (2010), respectively.

by lenders because of higher default risk; meanwhile, people with lower incomes are generally more price-sensitive or more price-elastic.

If we simply apply the BLP model to estimate mortgage demand and assume that borrowers face the same price for the same loan product, we will overestimate the effect of income in reducing borrowers' price sensitivities. In other words, we can overestimate price sensitivities of low-income borrowers and underestimate price sensitivities of high-income borrowers. Borrowers with higher incomes are more likely to purchase a financial product for two reasons: first, they have lower price sensitivities; second, they face lower prices because of their lower default risk. If we ignore the channel that customers with higher incomes face lower prices, we have to make these customers even less price-sensitive in the model to fit the data.

In this paper, we propose a two-step estimation for demand in financial-product markets in which different consumers face different prices for the same financial product and the price a consumer faces is related to the consumer's demographic characteristics. The minimum data requirement for implementing our method includes individual-level price data for a random sample of consumers and the demographic characteristics of these consumers (additional data compared to BLP);[2] data on characteristics and market shares of each product; and data on the distribution of demographic characteristics of the population in each market. In addition, we propose a method to correct biases arising from the selectivity that only the individual-specific price of the product chosen by an individual is observed by econometricians; individual-specific prices of products not chosen by the individual cannot be observed by econometricians while can be observed by the individual when she/he compares among products.

We conduct Monte Carlo simulations and show that when lenders conduct individual-risk-based pricing and hence the borrower-specific price for a financial product is decreasing in income, the traditional BLP model will overestimate the effect of income in reducing borrowers' price sensitivities. In contrast, our method (a modified BLP model) provides estimates that are closer to the true parameter.

Then, we apply our method to estimate the demand in the U.S. residential mortgage market. The estimate of the income effect on price sensitivities generated by the traditional BLP is 11.20% greater in magnitude than that generated by our method. Compared with our method, the traditional

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[2] While individual-level data for the entire population of customers are usually available in financial markets, a random sample is sufficient for implementing our method.

BLP method tends to overestimate price sensitivities of low-income borrowers and underestimate price sensitivities of high-income borrowers. Given that there are low-income borrowers than high-income borrowers in the population, the distribution of individual-specific price sensitivity recovered by the traditional BLP method has more high-price-sensitivity borrowers and fewer low-price-sensitivity borrowers than the distribution recovered by our method.

Finally, we conduct one of the possible further applications of our methodology: we quantify borrowers' willingness to pay for fintech based on the estimates of the structural parameters. We find that compared with the distribution of borrowers' willingness to pay for fintech recovered by our method, the distribution recovered by the traditional BLP method tends to have more borrowers with low willingness to pay and fewer borrowers with high willingness to pay.

We further quantify the value of face-to-face interaction for a first-time borrower to borrow from a non-fintech lender. The utility of the convenience or financial-literacy gains from face-to-face interaction for first-time borrowers is equivalent to that of paying from 125 to 200 bps lower interest rates. This result generates a managerial implication that it is substantially worthwhile for fintech lenders to consider posting videos on their websites showing mortgage borrowing steps in detail and addressing possible concerns to increase their market shares over first-time homebuyers. Fintech lenders may also consider investing in the provision of online video chat consultation services.

The rest of the paper is organized as follows. Section 2 discusses the related literature and our contributions. Section 3 describes our estimation methodology. Section 4 discusses the selectivity issue and proposes a feasible method to correct the selection bias. We present Monte Carlo simulation results in Appendix B. Section 5 addresses several concerns. Section 6 describes the data for the empirical application. Section 7 presents the empirical estimation results. In Section 8, we conduct one of the possible further applications of our methodology: quantifying borrowers' willingness to pay for fintech based on the structural estimates in Section 7. In Section 9, we discuss several other potential further applications of the methodology. We conclude in Section 10.

2. Literature and our contributions

Our study contributes to four strands of literature: demand estimation of financial products, BLP and its extensions, selectivity, and fintech.

2.1. Demand estimation of financial products

Many studies in the finance literature treat financial products as homogenous products and examine quantities demanded by households or firms (e.g., DeFusco and Paciorek, 2017 on mortgage demand). Other studies simply applied reduced-form estimation to analyze the take-up rates of a certain financial product without considering the substitution pattern across different financial products (e.g., Haurin et al., 2016 on the take-up rates of reverse mortgages). These frameworks cannot enable researchers to analyze typical issues in modern IO, including quantifying consumers' willingness to pay, market power, mergers, entry, pricing, regulation, and deregulation.

A few seminal studies building new bridges across finance and IO applied the traditional BLP model to estimate demand for financial products without allowing that different borrowers face different prices for the same financial product and that the price a borrower faces is related to its demographic characteristics (e.g., Whited et al., 2021; Wang et al., 2022; Buchak et al., 2018).[3] [4] This treatment may not be problematic for analyzing their research questions but can bias the estimates of borrowers' price sensitivities and hence bias welfare analyses for other potential research questions that are more intensively based on IO foundation and are of interest to both IO and financial economists, such as welfare analyses on bank mergers, bank branch deregulations, and governmental programs aiming to enhance financial access of low-income (financially disadvantaged) populations. In contrast, our method can be applied to analyze these research questions.

[3] Note that Whited et al. (2021), Wang et al. (2022), and Egan et al. (2017) also used the traditional BLP model to estimate the demand of depositors. Depositors in a market usually face the same interest rate for the same deposit product; thus, using the traditional BLP model is sufficient to estimate the demand for deposits.

[4] Koijen and Yogo (2019) and Gabaix et al. (2022) estimated characteristics-based demand models to match stock holdings by institutions and households for developing a new asset pricing model. First, their characteristics-based demand model is not the BLP-type: in the BLP model, individuals make discrete product choices, which are aggregated to match continuous market shares; in contrast, in the characteristics-based demand models in Koijen and Yogo (2019) and Gabaix et al. (2022), institutions or households make continuous choices on the shares of each stock in their portfolios, and portfolios held by each household or institution are observed. Second, in their studies, each investor faces the same price for the same stock at the same time point; in contrast, in our study, different borrowers face different prices for the same loan product at the same time point.

In financial-product industries, while individual-level data for the entire population of customers are usually available to researchers, the BLP framework utilizing market-level data still provides irreplaceable merits. One reason is that the BLP framework can compute the market-period-product-level unobserved demand shocks or product characteristics (ξ) through the contraction mapping (in which the uniqueness of the solution and the convergence are guaranteed by theory) instead of through nonlinear search in the optimization step (which is subject to the curse of dimensionalities). One group of existing studies conducted maximum likelihood estimations (MLE) based on individual-level observations without including ξ in borrowers' discrete choice models over financial products. However, omitting ξ will cause toward-zero biases in the estimates of price sensitivities (See the discussions in Berry et al. (1995), Nevo (2000), Petrin (2002), or Goolsbee and Petrin (2004)).[5]

A second group of existing studies first aggregated individual-level data to market-level data and then applied the BLP framework based on the market-level observations instead of the individual-level observations (e.g., Buchak et al., 2024 on the U.S. mortgage industry). Crawford et al. (2018) conducted MLE based on individual-level observations to estimate some structural parameters (e.g., the parameters governing default decisions), but the key parameters in their demand models (e.g., price sensitivities) were still estimated by the BLP framework based on the market-lever observations aggregated from individual-level data.

A third group (e.g., Benetton, 2021; Robles-Garcia, 2022; Benetton et al., 2021) conducted MLE based on individual-level observations to estimate demand parameters and included ξ, but they examined the U.K. mortgage industry and, given its national nature, they either treated the entire country as one market or defined a market either as one of the eleven regions in the U.K. and conducted MLE separately for each market.[6] Those studies did not include the individual-specific taste coefficients in their structural demand models: Benetton (2021) and Benetton et al. (2021), which defined a market as a region, estimated market-specific price sensitivities; Robles-Garcia (2022), which treated the entire country as a market, estimated semiannual-region-income-quartile-specific price sensitivities. In contrast, the BLP framework conducts the estimation simultaneously for all the markets because the cross-market variation in product choice sets and that in demographic distributions provide identification for some key parameters that govern the

[5] Petrin and Train (2003) provided an alternative approach based on control functions.
[6] Unlike the U.K. with eleven regions, the U.S. has almost 400 metropolitan statistical area (MSA) markets.

more reasonable substitution pattern across products, such as the parameters of observed demographic characteristics and the variance of unobserved demographic characteristics in determining individual-specific price sensitivities.[7] [8] [9]

Several studies discussed above also mentioned the selectivity issue that only individual-specific prices of products chosen by an individual can be observed by econometricians. They addressed this issue mainly based on specific institutional features that could limit the potential extent of this problem. Benetton (2021), Robles-Garcia (2022), and Benetton et al. (2021) studied the U.K. mortgage market, in which individual-based pricing or negotiation between lenders and borrowers is limited (unlike the U.S., Canada, and Continental Europe markets). Crawford et al. (2018) studied Italian small business lines of credit and provided evidence that the pricing is mainly based on "hard information." In contrast, our study provides formal econometric methods to resolve this issue. Applying our methods to the U.S. mortgage market, in which individual-based pricing or borrower-lender negotiation is prevailing, the key structural estimate generated by the method without correction for the selection bias is 6.66% greater than that generated by the method with the correction.

Another difference between Crawford et al. (2018) and our study is that they had a constant price sensitivity parameter instead of an individual-specific random coefficient that varies according to demographic characteristics because they examined firm loan markets instead of household loan markets. In contrast, in our study, demographic characteristics (e.g., income) affect both price sensitivities and individual prices; thus, to provide consistent estimates of structural

[7] For the difference between the unreasonable substitution pattern across products implied by the logit model without random coefficients and the more reasonable substitution pattern implied by the mixed multinomial logit model with random coefficients, see the discussions in Berry et al. (1995), Petrin (2002), Goolsbee and Petrin (2004), Train (1986), Hausman and Wise (1978).

[8] Based on individual-level observations in the Bolivian firm loan markets, Ioannidou et al. (2022) conducted MLE simultaneously for all the markets. Their model had individual-specific coefficients for price sensitivity but did not have ξ.

[9] Within the typical industrial organization literature on nonfinancial industries, Goolsbee and Petrin (2004) conducted MLE based on individual-level observations simultaneously for all the markets. To reduce the large dimension caused by the market-period-product-level fixed effects (to be used to recover ξ), given any candidate values of structural parameters, they solved such fixed effects market by market through matching the model-predicted market shares to observed market shares. Therefore, market-level observations were still used. Grieco et al. (2023) simultaneously included the likelihood function based on individual-level and market-level observations and the BLP moment function based on market-level observations in the objection function that was to be optimized w.r.t structural parameters.

parameters, we need to separately identify these two channels through which demographic characteristics influence the demand.[10]

2.2. BLP and extensions

The second strand of literature we contribute to is the literature that extended the traditional BLP model. Berry et al. (2004) derived the asymptotic property for the BLP model. Nevo (2000a, 2001) extended the BLP model to include brand dummies. Petrin (2002) and Berry et al. (2004) combined market share data with micro-level consumer survey data to estimate demand for differentiated products (known as micro-BLP). Conlon and Gortmaker (2023) provided a more general framework for incorporating many types of micro data (from summary statistics to full surveys of selected consumers) into the BLP framework. Berry and Haile (2020) examined the nonparametric identification of micro-BLP. Goolsbee and Petrin (2004) developed a likelihood-style BLP framework using both individual-level and market-level observations. Grieco et al. (2023) simultaneously included the likelihood function based on individual-level and market-level observations and the BLP moment function based on market-level observations in the objection function that was to be optimized w.r.t structural parameters.[11]

However, all these extensions and modifications are within nonfinancial-product markets with the underlying assumption that all consumers in a market face the same price for the same product. Our study extends the BLP model to financial-product markets, an essential economic sector with vast sales volumes and many important policy concerns. For example, mortgage originations in the U.S. during 2022 amounted to approximately 4 trillion dollars, compared to 0.86 trillion dollars for the U.S. motor vehicle and parts manufacturing gross output in the same year. Our method can also be applied to nonfinancial-product markets with price discrimination

[10] Ioannidou et al. (2022) studied the Bolivian firm loan markets and had firm-specific coefficients for price sensitivity that depend on the relationship dummy indicating an existing lending relationship between a bank and a firm, but the dummy did not present in their hedonic price prediction equation.

[11] To name a few other important extensions to BLP, Fan (2013) and Eizenberg (2014) extended the BLP model to allow for endogenous product attributes. Gentzkow (2007) and Berry and Jia (2010) allowed dependence between taste parameters. Goeree (2008) allowed different consumers to have different choice sets. Lu et al. (2019) allowed for nonparametric specification of preference heterogeneity. Gandhi et al. (2022) allowed for products with zero sales in the data. Dube et al. (2012) developed a new computational algorithm for implementing the BLP estimator (a mathematical program with equilibrium constraints, known as MPEC).

based on consumers' demographic characteristics that are correlated with their price elasticities of demand.

While our method requires both market-level data and a sample (or the entire population) of individual-level data, it differs from the existing micro-BLP models in the literature. First, in the existing micro-BLP models, all consumers in a market face the same price for the same product; prices are correlated with only product characteristics and market-level demand shocks (either observed or unobserved). In contrast, in our model, consumers in a market face different prices for the same product; individual-specific prices are correlated with not only product characteristics and market-level demand shocks but also individual-level demographic characteristics.

Second, estimating the existing micro-BLP models requires market share data and a sample of individual choice data; individual choice data are used for identification under weaker assumptions in theory and obtaining more precise estimates in practice. In contrast, our method requires market share data and a sample of individual price data; individual price data are used to identify and estimate the relationship between individual-specific prices and individual-level demographic characteristics that arises from borrower-risk-based pricing conducted by financial-product providers or price discrimination conducted by nonfinancial-product providers. Given that the cross-market variation in products' market shares and that in demographic distributions have already been used to identify how demographic characteristics influence individual-specific price sensitivities, we need individual-level price data to identify how demographic characteristics influence individual-specific prices.

To our best knowledge, the paper most closely related to our study is D'Haultfœuille et al. (2019). We differ from that paper mainly in requirements on data availabilities, assumptions for identification and computation, and hence applicable empirical settings. First, D'Haultfœuille et al. (2019) assumed that consumers can be divided into subgroups such that consumers in a subgroup have the same demographic characteristics observable to sellers and pay the same price for a product; econometricians observe subgroup-product-specific market shares but not subgroup-product-specific prices, which will be recovered in a contraction mapping in the estimation process. In contrast, our method does not involve this kind of subgroups; in financial product markets, borrowers with the same demographic characteristics commonly observable to all lenders (hard information, e.g., FICO) may not pay the same interest rate for a product of a lender because the lender may possess soft information for some of these borrowers but not for the others; moreover, different lenders may possess different levels of soft information of the same borrower. Our method requires a sample of individual-level data with individual-level prices and demographic characteristics, which is widely available in many financial industries, such as the residential mortgage industry. Because our method leverages observed individual-specific prices rather than recovers them from a contraction mapping, our method needs to handle the selectivity issue that individual-specific prices of only products chosen by individuals are observed.[12]

The second difference is that D'Haultfœuille et al. (2019) assumed that the marginal cost of a product is identical for all buyers. In contrast, our method does not need this assumption for identification, given the fact that the lending costs for borrowers for mortgages with the same loan characteristics can be different because borrowers differ in default risk.[13]

The third difference is that, to guarantee that the iteration for solving the mean utility level is a contraction mapping, D'Haultfœuille et al.'s (2019) method requires that the heterogeneity in consumer price sensitivity is not large. In contrast, our method does not have this requirement, and one advantage of our method over the traditional BLP is that our method will not overestimate the heterogeneity (income effect) on borrower price sensitivity in financial-product industries.

To summarize, D'Haultfœuille et al.'s (2019) method applies to nonfinancial industries with price discrimination based on consumers' demographic characteristics correlated with their price sensitivities (e.g., the automobile industry examined in that paper), whereas our method applies to financial industries with either risk-based pricing or price discrimination. Our method can also be applied to nonfinancial industries with price discrimination if individual-level price data for a random sample of consumers are available (see discussions in Section 9.4).

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[12] In the supplementary material of D'Haultfœuille et al. (2019), the authors extended their initial setup to the case where only aggregate market shares of products, rather than market shares of subgroups, are observed. It requires an additional assumption for identification: consumers are homogeneous inside each subgroup and only differ in their product-specific terms (usually supposed to be i.i.d. and extreme-value distributed). In contrast, our method does not need this assumption. The random coefficients in consumers' utility functions can have the unobserved heterogeneity component, such as v_i in equation (3.3) in Section 3.

[13] In the supplementary material of D'Haultfœuille et al. (2019), the authors extended their initial setup to the case where the marginal cost of a product can vary across subgroups of consumers. It requires an additional assumption for identification: all the cost shifters are observed by econometricians. In contrast, our method does not need this assumption. In the finance literature, while the FICO score is an important observed factor affecting borrowers' default risk, unobserved borrower characteristics matter for borrowers' default risk and hence for the lending costs.

2.3. Selectivity

As we provide a method to correct biases coming from selectivity that individual-specific prices of only products chosen by individuals are observed, our work also contributes to the literature on selectivity. The selectivity issue that our study handles is fundamentally different from the Heckman-type selectivity, in which individuals are selected into markets by their decisions, such as work participation (Heckman, 1974, 1976, and 1979) and insurance take-up (Einav et al., 2010; Einav and Finkelstein, 2011), and data for individuals not selected into markets are unobserved. In contrast, the selectivity in our study results from individuals' discrete choices in the market conditional on that these individuals have already been in the market. Choice-individual-specific information (individually customized prices for a product) is unobserved by econometricians for options (products) not chosen by an individual.

2.4. Fintech

The fourth strand of literature we contribute to is the literature on fintech (see Philippon, 2016 for an overview). Fuster et al. (2019) found that fintech lenders process mortgage applications 20% faster than non-fintech lenders. Buchak et al. (2018) found that fintech lenders charge higher interest rates than non-fintech lenders.[14] In contrast, we recover the distribution of borrowers' willingness to pay for fintech based on the structural estimates generated by our method. The results indicate that the positive premium charged by fintech lenders should possibly be explained by that fintech lenders exploit the relatively small portion of borrowers who have strong preferences for application convenience and speed and hence have a positive willingness to pay for fintech. The results can help fintech lenders develop pricing strategies and help traditional non-fintech lenders evaluate the competition they face from fintech lenders.

Given that Buchak et al. (2018) and Fuster et al. (2019) found reduced-form evidence that first-time homebuyers are less likely to choose fintech lenders, we use our structural estimation framework to quantify the value of face-to-face interaction for a first-time borrower to borrow from a non-fintech lender. The utility of the convenience or financial-literacy gains from face-to-

[14] There are multiple other studies on fintech. For example, Bartlett et al. (2018) found that the entry of fintech lenders can alleviate race-based discrimination in mortgage markets. Ma and Zhang (2022) studied the wealth management product market in China; they found that products sold at bank physical branches offer lower promised yields than products sold online and attributed this positive premium to compensating the inconvenience of offline purchasing for individual investors.

face interaction for first-time borrowers is equivalent to that of paying from 125 to 200 bps lower interest rates. This result generates a managerial implication that it is substantially worthwhile for fintech lenders to consider posting videos on their websites showing mortgage borrowing steps in detail and addressing possible concerns to increase their market shares over first-time homebuyers. Fintech lenders may also consider investing in the provision of online video chat consultation services.

3. Methodology

3.1. Model

Suppose that there are M city markets with T periods. The number of consumers in market m in period t is I_{mt}. There are J differentiated products and K firms.

The price (interest rate) of product j for consumer (borrower) i depends on macroeconomic conditions A_t (e.g., the baseline interest rate, the credit spread, and the term spread), market characteristics B_m, firm (lender) characteristics C_k, product characteristics X_j and borrower i's observable demographic characteristics D_i, shown as follows:

$$p_{ijmt} = p_j(A_t, B_m, C_k, X_j) + \theta_t D_i + \eta_{ijmt} \qquad (3.1)$$

η_{ijmt} in equation (3.1) is the unobserved factor affecting prices and is independent across i, j, m, and t. Assume that for period t, $\eta_{ijmt} \sim N(0, \sigma_{\eta_t}^2)$. If D_i is borrower i's income level, then $\theta_t < 0$; the reason is that borrowers with higher income levels have lower default risk and are thus charged lower prices (interest rates) by lenders.[15]

The utility obtained by borrower i in market m in period t from choosing product j is:

$$u_{ijmt} = -\alpha_i p_{ijmt} + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt} \qquad (3.2)$$

The borrower-specific price sensitivity α_i is determined as follows:

$$\alpha_i = \alpha_0 - \gamma D_i + v_i \qquad (3.3)$$

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[15] In equation (3.1), we write $\theta_t D_i$ separate from $p_j(A_t, B_m, C_k, X_j)$ for notation convenience given that our focus is the effect of D_i. Theoretically, our method can allow a more general form of equation (3.1) as

$$p_{ijmt} = p_j(A_t, B_m, C_k, X_j, D_i) + \eta_{ijmt}$$

where $p_j(.)$ is an unknown function to be recovered. We have to require η_{ijmt} to be separate from $p_j(A_t, B_m, C_k, X_j, D_i)$ instead of standing inside $p_j(A_t, B_m, C_k, X_j, D_i, \eta_{ijmt})$ because the theorems developed in Section 4 for selectivity bias correction rely on the separability of η_{ijmt}.

α_0 is constant across individual borrowers. $v_i \sim N(0, \sigma_v^2)$, which captures the effect of borrower i' unobservable demographic characteristics on her price sensitivity. If D_i is borrower i's income level, then $\gamma > 0$; the reason is that borrowers with higher income levels should be less sensitive to price increases. ξ_{jmt} represents unobserved (by econometricians) product characteristics, including clauses specified in the loan contract but not observed by econometricians, average branch characteristics in the location (e.g., proximity and clerk attitude), average processing time, location-time-specific demand shocks, etc.

In equation (3.2), ε_{ijmt} is borrower i's idiosyncratic taste for product j. Assume that ε_{ijmt} follows a Type I extreme-value distribution. Then, the probability for borrower i to choose product j is

$$Prob\{y_{ijmt} = 1\} = \frac{exp(-\alpha_i p_{ijmt} + \beta X_j + \xi_{jmt})}{1 + \sum_{j'} exp(-\alpha_i p_{ij'mt} + \beta X_{j'} + \xi_{j'mt})} \tag{3.4}$$

Denote the joint cumulative distribution function of (D_i, v_i) for the population in market m in period t as $F_{mt}(D_i, v_i)$. The market share of product j in market m in period t is:

$$s_{jmt} = \int \frac{exp(-\alpha_i p_{ijmt} + \beta X_j + \xi_{jmt})}{1 + \sum_{j'} exp(-\alpha_i p_{ij'mt} + \beta X_{j'} + \xi_{j'mt})} dF_{mt}(D_i, v_i) \tag{3.5}$$

3.2. Estimation procedures

Suppose that we observe the individual-level price data p_{ijmt} for a random sample of borrowers, as well as the demographic characteristics (D_i) of these borrowers; we also observe the market share of each product j in market m in period t (S_{jmt}); we do not observe all the relevant macroeconomic variables A_t, market characteristics B_m, and lender characteristics C_k in equation (3.1); we also do not know the functional form of $p_j(.)$ in equation (3.1).

Our estimation procedure has two steps. In the first step, for each period t, we use the individual-level price data (either the entire population or a random sample, depending on the availability) to fit the following hedonic pricing equation:

$$p_{ijmt} = \left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt} \tag{3.6}$$

In equation (3.6), $\underline{\overline{X}}_{jmt}$ includes the metropolitan statistical area (MSA) fixed effect, the lender fixed effect, and product characteristics such as the term category and LTV category; borrower demographic characteristics D_i include the borrower's FICO and income. Θ_t are the parameters to be estimated. From the regression for period t, we obtain the estimates $\widehat{\Theta}_t$ and $\hat{\sigma}^2_{\eta_t}$ (the variance of the residual).[16] Then, for each borrower in the population with demographic characteristics D_i, we can predict the average price (conditional on the borrower's demographic characteristics) the borrower needs to pay if purchasing product j in period t by plugging in the product characteristics $\underline{\overline{X}}_{jmt}$, the borrower's demographic characteristics D_i, and $\widehat{\Theta}_t$ obtained from the first-step estimation into the following equation:

$$\hat{p}_{ijmt} = \left(\underline{\overline{X}}_{jmt}, D_i\right)\widehat{\Theta}_t \tag{3.7}$$

In the second step, given the unobserved product characteristics $\xi_{mt} = \left\{\xi_{jmt}\right\}_{j\in J}$ and the numerical values for the structural parameters $\{\alpha_0, \gamma, \sigma_v, \beta\}$, we can simulate product market shares as follows:

$$s_{jmt}(\xi_{mt}; \alpha_0, \gamma, \sigma_v, \beta) = \frac{1}{ns}\sum_{i=1}^{ns} \frac{exp\left(-\alpha_i(\hat{p}_{ijmt} + \tilde{\eta}_{ij}) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i(\hat{p}_{ij'mt} + \tilde{\eta}_{ij'}) + \beta X_{j'} + \xi_{j'}\right.} \tag{3.8}$$

To simulate $s_{jmt}(\xi_{mt}; \alpha_0, \gamma, \sigma_v, \beta)$ in equation (3.8), we randomly draw D_i, v_i, and $\tilde{\eta}_{ij}$, $i = 1, \ldots, ns$, from $F_{D,mt}(D)$, $N(0, \sigma_v^2)$, and $N(0, \hat{\sigma}^2_{\eta_t})$, respectively. α_i in equation (3.8) is affected by D_i and v_i through equation (3.3). $\hat{p}_{ijmt}$ in equation (3.8) is affected by D_i through equation (3.7).

The reason why we use the predicted individual-product-specific prices $\hat{p}_{ijmt}$ rather than the original prices p_{ijmt} observed in the data to construct the market shares by equation (3.8) is that, for individual i, we only observe the price of the product chosen by this individual, but we also need the underlying individual-specific prices of products not chosen by this individual to construct the market shares.[17]

[16] Theoretically, our method can allow a more flexible form for equation (3.6) as

$$p_{ijmt} = p_j(A_t, B_m, C_k, X_j, D_i) + \eta_{ijmt}$$

and estimate $p_j(.)$ by nonparametric or semiparametric methods. Suggested methods include average derivative estimators for single-index models if the number of regressors is not too large (see Stoker (1986) and Hardle and Stoker (1989) for the case without discrete independent variables and Horowitz and Hardle (1996) for the case with discrete ones). Given that the number of independent variables is large, we employ linear regressions instead to reduce the computation burden and instability. As we include many fixed effects and run regressions separately for each month, our linear specification still has substantial flexibility (see discussions in Section 5.3).

[17] Given that our method requires individual-specific price data for a random sample of consumers rather than the entire population, if an individual randomly drawn from $F_{D,mt}(D)$ to construct the simulated

Let $s_{mt}(\xi_{mt};\alpha_0,\gamma,\sigma_v,\beta) = \{s_{jmt}(\xi_{mt};\alpha_0,\gamma,\sigma_v,\beta)\}_{j\in J}$ (simulated market shares) and $S_{mt} = \{S_{jmt}\}_{j\in J}$ (observed market shares in the data). Given a set of values for the parameters $\{\alpha_0,\gamma,\sigma_v,\beta\}$, the unobserved product characteristics ξ_{mt} can be solved from the following implicit system of equations:

$$s_{mt}(\xi_{mt};\alpha_0,\gamma,\sigma_v,\beta) = S_{mt} \tag{3.9}$$

Based on the contraction mapping suggested by BLP, ξ_{mt} can be numerically solved by the following iteration:

$$\xi_{mt}^{h+1} = \xi_{mt}^{h} + lnS_{mt} - lns_{mt}\left(\xi_{mt}^{h};\alpha_0,\gamma,\sigma_v,\beta\right) \tag{3.10}$$

Stack all the ξ_{jmt}s as a column vector ξ with L elements. Denote all the instrumental variables as matrix Z, with each column as one instrumental variable. Let $g_l(\alpha_0,\gamma,\sigma_v,\beta) = Z_l{}'\xi_l$, where Z_l is the lth row of matrix Z and ξ_l is the lth element of column vector ξ. Denote the moment conditions as:

$$g(\alpha_0,\gamma,\sigma_v,\beta) = \frac{1}{L}\sum_{l=1}^{L} g_l(\alpha_0,\gamma,\sigma_v,\beta) = \frac{1}{L}Z'\xi \tag{3.11}$$

The GMM estimate of the structural parameters can be obtained by solving the following minimization problem:

$$\min_{\{\alpha_0,\gamma,\sigma_v,\beta\}} g'Wg \tag{3.12}$$

W in equation (3.12) is the weighting matrix.[18]

3.3. Comparison with traditional BLP estimators

Regarding the demand for differentiated financial products, traditional BLP estimators will generate upward biases for the effect of income on a customer's price sensitivity.

Based on equations (3.2) and (3.3) in our model, the utility of customer i from purchasing product j in location m in period t is:

$$u_{ijmt} = -(\alpha_0 - \gamma D_i + v_i)p_{ijmt} + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt} \tag{3.13}$$

In contrast, based on the traditional BLP model, the utility is:

$$u_{ijmt} = -(\alpha_0 - \gamma_0 D_i + v_i)\bar{p}_{jmt} + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt} \tag{3.14}$$

market share in equation (3.8) is not in the price sample, we need to predict the individual-specific prices of every product in the choice set regardless of whether the product is chosen or not chosen by the individual.

[18] Following Nevo (2000a, 2001), the weighting matrix can be Φ^{-1}, where Φ is a consistent estimate of $E[Z'\xi\xi'Z]$.

where $\bar{p}_{jmt}$ is the average price paid by customers (or the listed price) for product j in location m in period t instead of the individual-specific price p_{ijmt}.

Suppose that the income (D_i) of a customer increases. The component $(\alpha_0 - \gamma D_i + v_i)p_{ijmt}$ in the customer's utility function (the disutility from paying prices) will decrease through the decrease in $\alpha_0 - \gamma D_i + v_i$ and the decrease in the individual-specific price p_{ijmt} (borrowers with higher incomes have lower risk and hence are charged lower interest rates). If we replace the individual-specific price p_{ijmt} with the average price $\bar{p}_{jmt}$ (as shown in equation (3.14)), the disutility from paying prices will become $(\alpha_0 - \gamma_0 D_i + v_i)\bar{p}_{jmt}$. When the income ($D_i$) increases, $\bar{p}_{jmt}$ does not change. To make $(\alpha_0 - \gamma_0 D_i + v_i)\bar{p}_{jmt}$ decrease by the same amount as $(\alpha_0 - \gamma D_i + v_i)p_{ijmt}$, γ_0 has to be greater in magnitude than γ. Intuitively, customers with higher incomes are more likely to purchase a financial product for two reasons: first, they have lower price sensitivities; second, they face lower prices. If we ignore the channel that customers with higher incomes face lower prices, we have to make these customers even less price-sensitive to fit the data.

3.4. Variance of the estimator

Following the spirit of Berry et al. (1995) and Berry et al. (2004), the asymptotic covariance matrix of the estimators can be calculated as:

$$(G'WG)^{-1}G'W\left(\sum_{i=1}^{3} V_i\right)WG(G'WG)^{-1} \tag{3.15}$$

where

$$G = \frac{\partial g}{\partial (.)'} \tag{3.16}$$

and

$$V_1 = \frac{1}{L}\sum\nolimits_{l=1}^{L}\sum\nolimits_{\tilde{l}=1}^{L} g_l g_{\tilde{l}}' \tag{3.17}$$

V_2 in equation (3.15) arises from the simulation process that generates the difference between the empirical distribution of demographic characteristics of randomly selected borrowers and the distribution of the entire population. Following Berry et al. (1995) and Berry et al. (2004), V_2 can be estimated via a Monte Carlo procedure. Specifically, we randomly draw a sample of borrowers to form an empirical distribution of demographic characteristics independently times. For each of these samples, we calculate the vector of moment conditions (3.11) and use the empirical variance of these moment conditions as our estimate of V_2.[19]

The difference between our method and the traditional BLP in the covariance matrix calculation is that our method has an additional component in the covariance matrix, V_3, which arises from the simulation process that generates the difference between the predicted individual price $\hat{p}_{ijmt} + \eta_{ijmt}$ and the true individual price p_{ijmt}. The variance of the difference comes from two sources. The first source is the variance of the estimator $\widehat{\Theta}_t$, which leads to the variance of the predicted average price $\hat{p}_{ijmt}$ conditional on demographic characteristics D_i, i.e.,

$$\begin{aligned} var(\hat{p}_{ijmt}) &= \left(\underline{\overline{X}}_{jmt}, D_i\right) var\left(\widehat{\Theta}_{jt}\right) \left(\underline{\overline{X}}_{jmt}, D_i\right)' \\ &= \hat{\sigma}^2_{\eta_t} \left(\underline{\overline{X}}_{jmt}, D_i\right) \left[\left(\underline{\overline{X}}_t, D_t\right)\left(\underline{\overline{X}}_t, D_t\right)'\right]^{-1} \left(\underline{\overline{X}}_{jmt}, D_i\right)' \end{aligned} \tag{3.18}$$

where $\left(\underline{\overline{X}}_t, D_t\right)$ include all $\left(\underline{\overline{X}}_{jmt}, D_i\right)$ in period t. The second source is the variance of the residual η_{ijmt} in period t, $\hat{\sigma}^2_{\eta_t}$.

For each borrower i and each product j in the borrower's choice set, we randomly draw a $\zeta_1 \sim N\left(0, var\left(\hat{p}_{ijmt}\right)\right)$ and a $\zeta_2 \sim N\left(0, \hat{\sigma}^2_{\eta_t}\right)$. We use $\hat{p}_{ijmt} + \zeta_1 + \zeta_2$ as an approximation of the true individual-specific price p_{ijmt} and calculate the vector of moment conditions (3.11). We repeat this process independently 100 times and use the empirical variance of these moment conditions as our estimate of V_3.

3.5. Instrumental variables

The first set of instrumental variables are X_j themselves. The linear structural parameters β are identified by these instrumental variables.

The second set of instrumental variables are supply-side cost shifters, which can help identify the nonlinear structural parameters $\{\alpha_0, \gamma, \sigma_v\}$. Lenders face two major costs: funding costs and operating costs. Lenders' funding costs will affect lenders' lending interest rates p_{ijmt}. Two types

[19] Berry et al. (1995) and Berry et al. (2004) had another component in the theoretical covariance matrix that arises from the consumer sampling process. However, as discussed in BLP, given that the number of consumers in the U.S. economy is very large, the magnitude of this component is negligible in empirical practices.

of factors can shift lenders' funding costs: common factors and lender-specific factors. Common factors include the baseline interest rate (1-year treasury constant maturity yield), the credit spread, and the term spread in period t, which are determined by macroeconomic conditions. Given the current macroeconomic condition, the credit spread (Baa corporate bond yield minus 10-year treasury constant maturity yield) measures how much premium investors require for taking additional default risk; the term spread (10-year treasury constant maturity yield minus 1-year treasury constant maturity) measures how much premium investors require for additional investment horizon. These common funding-cost shifters are uncorrelated with the unobserved part of borrowers' preferences (ξ_{jmt}). Furthermore, the following two types of variables can also serve as instrumental variables: the credit spread multiplied by the product characteristics that are related to a product's default risk (e.g., the LTV category); the term spread multiplied by the maturity. Through the multiplication, we expand the instrumental variables from the monthly level to the month-product level.

Lender-specific funding-cost shifters include lenders' funding structures. The funding of a lender mainly comes from two sources: deposits and wholesale funding (such as interbank borrowing). For lenders, wholesale funding is more costly than deposits. Therefore, the percentage of a lender's funding that comes from deposits is correlated with the lender's overall funding costs. Data on lenders' funding structures are available in lenders' financial reports. However, lenders' funding structures may not satisfy the exogeneity requirement for instrumental variables. Lenders' ability to absorb deposits partially depends on branch distribution, and branch distribution can affect borrowers' preferences. Therefore, in the empirical section, we do not use lenders' funding structure as an instrumental variable.

Lenders' operating costs include wage costs and noninterest expenses related to the use of fixed assets, which are available in lenders' financial reports. However, for the mortgage market, lenders' operating costs are priced into one-time upfront fees (including mortgage origination fees), while lenders' funding costs are priced into mortgage interest rates. As loan amounts of mortgages are large and loan terms of mortgages are long, one-time upfront fees are ignorable compared to periodical interest payments. We focus on the market of mortgages for house purchases instead of for refinance because upfront fees can affect the decisions of whether to refinance. Therefore, in our empirical session, the price in borrowers' utility function includes only interest rates and does not include upfront fees (usually unobservable in mortgage data); and we do not use operating costs as instrumental variables for prices. For other financial products that involve smaller loan amounts and shorter loan terms (e.g., automobile loans, P2P lending, payday loans, and consumption loans), upfront fees are not ignorable and operating costs can serve as instrumental variables.

A third set of potential instrumental variables rely on the assumption that the location of products in the characteristics space is exogenous. Based on Berry et al. (1995), for product i offered by a firm, researchers can use the sums of the values of the same characteristics of other products offered by that firm and the sums of the values of the same characteristics of products offered by other firms as the instrumental variables. While we employ this type of instrumental variables in the Monte Carlo simulation (see Appendix B), we do not use this type of instrumental variables in the empirical section for mortgages (Section 7). The first reason is that many characteristic variables are dummy variables (e.g., the indicator for the 30-year loan term); thus, this type of instrumental variables lack variation. The second reason is that, theoretically, each lender can provide the full set of financial products in the characteristic space, which is different from nonfinancial products (e.g., automobiles).

A fourth set of potential instrumental variables can be constructed by exploiting the panel structure of the data. Based on Nevo (2000b), if researchers can observe many local markets for many periods, brand-specific dummy variables can be included in X_j. Accordingly, the error term ξ_{jmt} should be replaced by $\Delta\xi_{jmt} = \xi_{jmt} - \bar{\xi}_{jm}$, where $\bar{\xi}_{jm}$ is the mean of ξ_{imt} within market m. If researchers assume that the city-specific valuations of the product, $\Delta\xi_{jmt}$, are independent across cities but are allowed to be correlated within a city over time, prices of product j in other cities could be valid instrumental variables. On the one hand, prices in other cities are correlated with the price in city m because of the common cost; on the other hand, prices in other cities are uncorrelated with the market-specific valuations of the product in city m. Panel data are available for many financial products.

All the instrumental variables discussed above are at the market-month-product level at most, and they are all for the second-stage estimation. One possible question is whether we need individual-level instrumental variables for the regressions of prices in the first stage. In fact, the purpose of the first-stage estimation for the hedonic price equations is to provide accurate predictions for the individual-specific prices to be plugged into the second-stage estimation. There is no attempt to interpret the coefficients generated from the first-stage estimation. Therefore, the concern in the first-stage estimation is the fit of the model specification rather than the consistency of the parameter estimates, and thereby we do not need individual-level instrumental variables at this stage (see more discussion in Section 5.2).

3.6. Identification

Borrowers' demographic characteristics (such as income) play two roles in the model. First, they affect individual-specific prices through θ in equation (3.1) due to individual-risk-based pricing conducted by lenders. Second, they affect borrowers' price sensitivities (preferences) through γ in equation (3.3).

The first effect (θ) is estimated in the first step and can be identified from the sample of individual price data (by the variation in individual specific prices and the variation in individual demographic characteristics). The second effect (γ) is estimated in the second step and can be identified by the variation in the distribution of demographic characteristics (such as income) across different markets. Simply speaking, if γ is large, a market with a slightly higher average income level tends to have a dramatically smaller market share for the outside option because borrowers in this market are substantially less price-sensitive on average.

The identification of other parameters in the model ($\{\beta, \alpha_0, \sigma_v\}$) is the same as that in the traditional BLP.

4. Selectivity and bias correction

4.1. A selectivity issue and the resulting bias

One concern is that when we estimate the hedonic pricing equation (3.6) in the first step, we only observe the individual-specific price of the product chosen by an individual and do not observe individual-specific prices of products not chosen by the individual, whereas in the second step, we use the estimated equation (3.6) to predict individual-specific prices of not only the product chosen by an individual but also products not chosen by the individual. This can cause a selectivity issue. In this section, we develop a method to correct biases caused by this selectivity issue.

On the one hand, many reasons for which a borrower chooses a certain product are unrelated to prices, including heterogeneous preferences over product characteristics and idiosyncratic tastes (ε_{ijmt}). The corresponding selectivity will not bias the estimation of the reduced-form parameter in the hedonic pricing equation (3.6) in the first step.

On the other hand, one possible reason for a product being chosen by an individual is that the residual η_{ijmt} in equation (3.6) for the product-individual combination (observed by the individual but not researchers) is low. For example, for a product chosen by a borrower, the borrower uses other financial services provided by the same lender; thus, the lender knows more soft information than other lenders about the borrower and hence offers a lower interest rate, or the lender simply offers discounts for old customers. For another example, the borrower happens to obtain a low negotiated price with the lender and then chooses the product. Consequently, the individual-specific prices of products chosen by borrowers tend to have lower residuals (η_{ijmt}) than the individual-specific prices of products not chosen by borrowers.

Therefore, if equation (3.6) represents the hedonic pricing equation for all the products faced by borrower i, the mean of p_{ijmt} conditional on product j being chosen by borrower i (denoted as $y_{ijmt} = 1$) is:

$$E[p_{ijmt}|y_{ijmt} = 1] = \left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + E[\eta_{ijmt}|y_{ijmt} = 1] \tag{4.1}$$

where

$$\begin{aligned}
&E[\eta_{ijmt}|y_{ijmt} = 1] \\
&= \int \eta_{ijmt} \frac{\int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{i(j-1)mt}, d\eta_{i(j+1)mt}, \dots, d\eta_{iJmt}}{\int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{iJmt}} d\eta_{ijmt} \\
&= \frac{\int \dots \int \eta_{ijmt} \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{iJmt}}{\int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{\overline{X}}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{iJmt}} = \frac{\widetilde{\mathfrak{H}}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)}{\underset{\sim}{\mathfrak{H}}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)} \\
&= \mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)
\end{aligned} \tag{4.2}$$

$h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)$ in equation (4.2) is the probability density function of the standard normal distribution; $\mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$ is a function of the characteristics of all products in the market and the demographic characteristics of borrower i; $\widetilde{\mathfrak{H}}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$ and $\underset{\sim}{\mathfrak{H}}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$ are the numerator and denominator, respectively, in $\mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$.

With only individual-specific prices of products chosen by individuals observed, what we estimate is equation (4.1) rather than equation (3.6). If $E\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right] = 0$, the selectivity issue will not bias the OLS estimates of Θ_t.

However, in general, $E[\eta_{ijmt}|y_{ijmt} = 1]$ is negative. Given that the probability density function $h\left(\frac{\eta_{ijmt}}{\sigma_{\eta_t}}\right)$ is axially symmetric at $\eta_{ijmt} = 0$, the reason why $E[\eta_{ijmt}|y_{ijmt} = 1]$ is negative is that the probability of product j being chosen by borrower i is axially asymmetric at $\eta_{ijmt} = 0$: $P\left(y_{ijmt} = 1\right)$ is high when $\eta_{ijmt} < 0$; $P\left(y_{ijmt} = 1\right)$ is low when $\eta_{ijmt} > 0$.

4.2. Two difficulties

Given that we only observe the price of the product chosen by borrower i, the OLS regression of p_{ijmt} will generate biased estimates for Θ_t in equation (3.6). Ideally, if we know the numeric value of $\mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$, we should run the following regression instead:

$$p_{ijmt} - \mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right) = \left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \epsilon_{ijmt} \tag{4.3}$$

Unfortunately, calculating $\mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$ requires the values of structural parameters in the utility function, which are to be estimated in the second step. However, based on equation (4.3), we can form the following moment conditions:

$$\begin{aligned} E\left[\left(p_{ijmt} - \mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right) - \left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t\right)\left(\underline{\overline{X}}_{jmt}, D_i\right)'\right] \\ = E\left[\epsilon_{ijmt}\left(\underline{\overline{X}}_{jmt}, D_i\right)'\right] = 0 \end{aligned} \tag{4.4}$$

We can combine the moment conditions in equation (4.4) and the BLP moments in equation (3.11) and simultaneously estimate Θ and the structural parameters $(\alpha_0, \gamma, \sigma_v, \beta)$ by GMM.

However, this simultaneous estimation method faces two difficulties. First, the integrals in equation (4.2) do not have close forms; thus, we need to calculate the numeric integrals by simulation, which is feasible if there are only several products in the market but will incur too much computation burden if the number of products is large (the empirical setting in this paper has 105 products).

The second difficulty is that Θ in the hedonic pricing equation (3.6) has a large number of dimensions, such as MSA fixed effects and period fixed effects. If we estimate Θ together with the structural parameters $(\alpha_0, \gamma, \sigma_v, \beta)$, we will face the curse of dimensionalities. While β can be linearly solved (as discussed in Nevo (2000b)), Θ cannot be linearly solved because it not only appears linearly in $\left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t$ in equation (4.4) but also nonlinearly affects $\mathfrak{H}\left(X_j, \underline{\overline{X}}_{mt}, D_i\right)$ in equation (4.4) in the way described in equation (4.2).

4.3. Feasible bias correction

Fortunately, the following two theorems help us resolve the two difficulties (see the proofs in Appendix A):

Theorem 1. As the number of products $J \to +\infty$,

$$\lim_{J \to +\infty} E\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right] = -\alpha_i \sigma_{\eta_t}^2 \tag{4.5}$$

Theorem 2. As the number of products $J \to +\infty$,

$$\lim_{J \to +\infty} Var\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right] = Var[\eta_{ijmt}] = \sigma_{\eta_t}^2 \tag{4.6}$$

Theorem 1 means that we do not need to numerically calculate the integrals in equation (4.2) for $E\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right]$. Asymptotically, $E\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right]$ is not related to product characteristics X_j and $\underline{\overline{X}}_{mt}$.

The asymptotic bias ($-\alpha_i \sigma_{\eta_t}^2$) implied from Theorem 1 is consistent with intuition: first, if $\sigma_{\eta_t}^2$ is large, a consumer will be more likely to be offered a product with an extremely low η_{ijmt} and hence choose the product; second, if α_i is large, a consumer cares more about prices relative to other product characteristics (X_j) and hence will be more likely to choose a product with a low η_{ijmt}.

Theorem 2 means that the selectivity issue (using only individual-specific prices of products chosen by individuals to estimate equation (3.6)) will not bias the estimates of the unconditional variance of residuals ($\sigma_{\eta_t}^2$).

If we know the numeric values of α_i and $\sigma_{\eta_t}^2$, we can run the following regression:

$$p_{ijmt} + \alpha_i \sigma_{\eta_t}^2 = \left(\underline{\overline{X}}_{jmt}, D_i\right)'\Theta_t + \epsilon_{ijmt} \tag{4.7}$$

and obtain a consistent estimate of Θ_t as follows:

$$\widetilde{\Theta}_t = \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{p_{ijmt} + \alpha_i \sigma_{\eta t}^2\}_i$$

$$= \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{p_{ijmt}\}_i$$

$$+ \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{\alpha_0 - \gamma D_i + v_i\}_i \sigma_{\eta t}^2$$

$$= \widehat{\Theta}_t + \alpha_0 \sigma_{\eta_t}^2 \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{1\}_i$$

$$- \gamma \sigma_{\eta_t}^2 \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{D_i\}_i$$

$$+ \sigma_{\eta_t}^2 \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{v_i\}_i$$

Because v_i is independent of $\left(\underline{\overline{X}}_{jmt}, D_i\right)$, the following equation also provides a consistent estimator of Θ_t:

$$\begin{aligned}\widetilde{\widetilde{\Theta}}_t = \widehat{\Theta}_t + \alpha_0 \sigma_{\eta_t}^2 \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{1\}_i \\ - \gamma \sigma_{\eta_t}^2 \left[\left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i\right]^{-1} \left\{\underline{\overline{X}}_{jmt}, D_i\right\}_i{}' \{D_i\}_i\end{aligned} \tag{4.8}$$

That is, to correct the bias of $\widehat{\Theta}_t$ due to selectivity, we need to add $\alpha_0 \sigma_{\eta_t}^2$ to the intercept in $\widehat{\Theta}_t$ and subtract $\gamma \sigma_{\eta_t}^2$ from the coefficient of D_i in $\widehat{\Theta}_t$. Furthermore, to consistently predict individual-specific prices, we can employ the following equation:

$$\tilde{p}_{ijmt} = \left(\underline{\overline{X}}_{jmt}, D_i\right) \widehat{\Theta}_t + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i = \hat{p}_{ijmt} + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i \tag{4.9}$$

Plugging $\tilde{p}_{ijmt} + \eta_{ij}$ instead of $\hat{p}_{ijmt} + \eta_{ij}$ as a proxy for p_{ijmt} into equation (3.8), we have the simulated market shares as follows:

$$\begin{aligned}& s_{jmt}(\xi_{mt}; \alpha_0, \gamma, \sigma_v, \beta) \\ &= \frac{1}{ns} \sum_{i=1}^{ns} \frac{exp\left(-\alpha_i(\hat{p}_{ijmt} + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i + \tilde{\eta}_{ij}) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i(\hat{p}_{ij'mt} + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i + \tilde{\eta}_{ij'}) + \beta X_{j'} + \xi_{j'mt}\right)}\end{aligned} \tag{4.10}$$

In equation (4.10), $\hat{p}_{ijmt}$ and $\sigma_{\eta_t}^2$ can be obtained from the OLS regression in the first step, α_0 and γ, together with other structural parameters, can be estimated by the GMM in the second step, and $\tilde{\eta}_{ij}$ is a random draw from $N(0, \hat{\sigma}_{\eta_t}^2)$.

To summarize, on the one hand, we no longer need to numerically calculate the integral in equation (4.2). On the other hand, we can separate the estimation of the large-dimension Θ_t as a linear regression in the first step from the nonlinear search for a limited number of structural parameters $(\alpha_0, \gamma, \sigma_v)$ in the second step and thereby avoid the curse of dimensionalities.

This asymptotic approximation for bias correction relies on the argument that the number of products $J \to +\infty$. Note that the asymptotic property of the traditional BLP estimator also relies on the argument that the number of products $J \to +\infty$ (see Berry et al., 1995; Berry et al., 2004). Therefore, our method does not impose additional assumptions on top of the traditional BLP. In the empirical settings of Berry et al. (1995), Nevo (2001), and our study, J equals 150, 25, and 105, respectively. Moreover, as the asymptotic price prediction bias is not related to product characteristics X_j, the correction is robust to misspecification of the utility functional form w.r.t. X_j (e.g., omitting higher-order terms of X_j).

5. Concerns

5.1. Do we need to correct biases differently for chosen and unchosen products?

One concern is that although as a proxy for an individual-specific price of a product (regardless of whether the product is chosen or not chosen by the individual), $\tilde{p}_{ijmt}$ corrects the bias caused by selectivity, it may suffer from an upward bias for a product chosen.[20]

However, in the simulation of the second-step estimation, we only need a consistent prediction for an individual-specific price of a product before whether the product is chosen or not is realized. Which product an individual eventually chooses does not matter because instead of using MLE based on individual choice data, our method employs GMM based on ξ inverted from market share data; essentially, our method is searching for parameters that make the structural model best fit market share data instead of individual choice data. As market shares are aggregated from simulated individual choice probabilities, only simulated individual choice probabilities instead of individual choices matter.

The reason why we need to correct the selection bias in the first-step estimation is that the first-step estimation is based on individual-level data and only prices of products chosen are observed; without the bias correction, we cannot have a consistent prediction for an individual-

[20] $\tilde{p}_{ijmt}$ cannot suffer from a downward bias for a product not chosen because

$$\lim_{j \to +\infty} \{E[\eta_{ijmt} | y_{ijmt} = 0] - E[\eta_{ijmt}]\} = 0$$

Given that there are an infinite number of products, $Prob(y_{ijmt} = 0) = 1$.

specific price of a product before whether the product is chosen or not is realized, which is to be used in the second-step estimation.

Moreover, in equation (4.10), we also have $\tilde{\eta}_{ij}$ randomly drawn from $N(0, \hat{\sigma}_{\eta_t}^2)$. Thus, for a simulated individual, the expectation of $\tilde{\eta}_{ij}$ conditional on product j being chosen is lower than the expectation of $\tilde{\eta}_{ij'}$ conditional on product j' not being chosen.[21]

5.2. Soft information

One possible concern is that lenders may know some soft information about borrowers, which is unobservable to econometricians. Consequently, the regression of equation (3.6) may incur the omitted-variable bias for Θ.

First of all, the purpose of the regression is to establish a hedonic pricing equation (correlation between the regressors and prices); by no means is it estimating the causal effect of the regressors on prices. As the value of $\widehat{\Theta}$ (the OLS estimator of Θ) can be affected by omitted soft information, $\hat{p}_{ijmt}$ captures not only the contribution from the observables but also the contribution from the part of soft information that comoves with the observables; the part of soft information that is independent of (orthogonal to) the observables is left in the residual $\hat{\eta}_{ij}$. For a price p_{ijmt} that is to be predicted, the orthogonal part of soft information is left in the gap between the true price and the predicted one with the bias being corrected, i.e., $p_{ijmt} - \left(\hat{p}_{ijmt} + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i\right)$, denoted as $\tilde{\tilde{\eta}}_{ij}$, which is unobservable given that p_{ijmt} is unobservable. Then, the utility function (3.2) can be written as

$$u_{ijmt} = -\alpha_i\left(\hat{p}_{ijmt} + \alpha_0 \sigma_{\eta_t}^2 - \gamma \sigma_{\eta_t}^2 D_i\right) + \beta X_j + \xi_{jmt} - \alpha_i \tilde{\tilde{\eta}}_{ij} + \varepsilon_{ijmt} \tag{5.1}$$

The idiosyncratic shock in equation (5.1) becomes $-\alpha_i \tilde{\tilde{\eta}}_{ij} + \varepsilon_{ijmt}$ instead of ε_{ijmt}. Given that $\tilde{\tilde{\eta}}_{ij}$ is uncorrelated with the observables and has a zero mean, $\alpha_i \tilde{\tilde{\eta}}_{ij}$ is uncorrelated with the

[21] Although we can construct consistent predictions of individual-specific prices to be plugged into the second-step estimation, they are still not the true prices p_{ijmt}. Thus, we may face the measurement error issue. However, in the situation where different consumers in a market face the same price of a product, according to Berry (1994), measurement error in product-level prices may not be a serious problem because product-level prices are already treated as endogenous variables and instrumented with variables that are uncorrelated with errors. Similarly, measurement error in individual-specific prices may not be a serious problem for the same reason. The evidence is that we obtain similar parameter estimates regardless of whether $\tilde{\eta}_{ij}$ randomly drawn from $N(0, \hat{\sigma}_{\eta_t}^2)$ is included in equation (4.10).

observables. Therefore, the existence of soft information will not cause the endogenous problem in the second step that idiosyncratic shocks are correlated with prices.

However, $\tilde{\tilde{\eta}}_{ij}$ may be correlated among products within the same lender because each lender uses the same soft information of a borrower to price all the products. Moreover, $\tilde{\tilde{\eta}}_{ij}$ may even be correlated across different lenders. Correspondingly, we can assume that the idiosyncratic shocks follow the nested logit distribution and allow them to be correlated among products within a lender or among all the options except the outside option.[22]

An alternative approach to handle the soft information issue is that if the individual-level price data are available for the entire population (usually available in financial markets) or for all the consumers drawn from the population when calculating the simulated market share in equation (4.10) instead of being available only for a generic random sample, we can employ $\hat{\eta}_{ij}$ obtained from the first step as a proxy for borrower i' soft information and substitute for $\tilde{\tilde{\eta}}_{ij}$ in equation (5.1). Then, the unobserved idiosyncratic shock becomes ε_{ijmt} and follows the logit distribution.[23] This approach has a similar spirit to control function methods.[24]

On top of this approach, we can further allow the unobserved demographic characteristic affecting price sensitivity (v_i in equation (3.3)) to be correlated with the soft information $\hat{\eta}_{ij}$ affecting prices and estimate the correlation. Assuming that $(\hat{\eta}_{ij}, v_i)$ follows a joint normal distribution $N(0,0,\hat{\sigma}^2_{\eta_t},\sigma^2_v,\rho)$, for each borrower i, given $\hat{\eta}_{ij}$, we can simulate v_i by the following:

$$v_i = \left(\rho \frac{\hat{\eta}_{ij}}{\hat{\sigma}_{\eta_t}} + \sqrt{1-\rho^2} e_{i2}\right)\sigma_v \tag{5.2}$$

where e_{i2} is randomly drawn from the standard normal distribution. Then, we plug the simulated v_i into equation (3.8) to obtain the simulated market shares. The variation in the empirical distributions of $\hat{\eta}_{ij}$ across markets provides identification for the correlation parameter ρ. Simply speaking, if ρ is highly positive, a market with a slightly higher average $\hat{\eta}_{ij}$ tends to have a much greater market share for the outside option because borrowers in this market tend to be much more price-sensitive on average due to the highly positive correlation between $\hat{\eta}_{ij}$ and v_i.

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[22] Our method is extendable to the case in which the idiosyncratic shocks follow the nested logit distribution. It is easy to show that Theorems 1 and 2 still hold under the nested logit assumption for the idiosyncratic shocks.

[23] We obtain similar parameter estimates using this method.

[24] See Wooldridge (2015) for a review of control function methods.

5.3. Hedonic price equations vs. lenders' pricing rules

By no means is the first-step estimation trying to explicitly and exactly recover lenders' pricing rules (strategies). The purpose of the first-step estimation is only to recover a reduced-form relationship between prices and observables with a satisfactory fit. Instead of explicitly modeling the consistency between the hedonic price equation and lenders' pricing rules based on the estimated consumer demand, we only need the reduced form of the hedonic price equation to be sufficiently flexible such that lenders' pricing rules can be implicitly embedded.[25]

To explicitly model firms' pricing strategy, we need a supply-side model with assumptions on the market structure and the game played by firms (e.g., the Bertrand-Nash game). A supply-side model is not necessary for the demand estimation using BLP, though it can improve the efficiency of the estimates of the demand-side parameters. Some empirical studies using BLP for demand estimation do not have a supply-side model because tractable classic assumptions for market structure may not be appropriate for their empirical settings (e.g., Jia (2022)).

Theoretically, our method can allow a more flexible form for the hedonic price equation (3.6) as

$$p_{ijmt} = p_j(A_t, B_m, C_k, X_j, D_i) + \eta_{ijmt}$$

$p_j(.)$ can be estimated by nonparametric or semiparametric methods in the first step. Given that the number of independent variables in our hedonic price equation is large, we employ linear regressions instead to reduce the computation burden and instability. As we include many fixed effects and run regressions separately for each month, our linear specification still has substantial flexibility. The time-varying MSA fixed effects can capture the market structure and competitiveness of an MSA market.

5.4. BLP based on market share data vs. MLE based on individual choice data

The availability of consumer-level data in financial industries is usually higher than that in non-financial industries. Meanwhile, consumer-level data are increasingly available for many non-

[25] The spirit is similar to that in the conditional choice model (CCP) developed by Hotz and Miller (1993). In the first step, a flexible reduced-form policy function of state variables is estimated. Because agents' decision rules in a dynamic choice setting are embedded in the policy function, the decision rules can be recovered in the second step based on the estimated policy function.

financial industries. Correspondingly, a natural question is whether BLP based on market share data (or micro BLP partially based on market share data) is dominated by maximum likelihood estimation (MLE) based on individual choice data if the latter data exist.

First of all, our method with the selection-bias correction is extendable to MLE using individual choice data. The estimation procedure still involves two steps. In the first step, we estimate hedonic pricing equations to predict individual-specific prices for products not chosen by individuals. In the second step, we plug the predicted prices with bias corrections into multinomial logit choice probabilities in the likelihood function instead of into simulated market shares in equation (4.10). We can employ full information maximum likelihood (FIML) or mixed distributions to deal with price endogeneity, prediction errors (assuming a distribution for prediction errors), and unobserved heterogenous consumer preferences (random coefficients).[26] For non-financial industries, we do not need to conduct the first step to predict prices of products not chosen by an individual, given that consumers face the same price for the same product.

One disadvantage of MLE using individual choice data is that it is difficult to conduct nonlinear search for a large number of unobserved market-period-product-level demand shocks (ξ_{jmt}) due to the curse of dimensionalities, while omitting ξ_{jmt} can bias the parameter estimates (see discussion in Nevo (2000)).[27] In contrast, the BLP framework solves ξ_{jmt} by the contraction mapping, and the uniqueness of the solution and the convergence are guaranteed by theory. We have already compared the irreplaceable merits of the BLP framework with a number of existing studies conducting MLE based on individual-level observations in Section 2.2.

Another disadvantage is that the linear parameters β have to be searched nonlinearly in MLE together with nonlinear parameters, which can also trap econometricians into the curse of dimensionalities. Especially when FIML or mixed distributions are employed, the objective function is not regular, making the global maximum extremely difficult to find in a high-dimension space.

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[26] See Fox et al. (2012) for the identification requirements on the random coefficient multinomial logit model.

[27] To embed unobserved product characteristics ξ_{jmt} into MLE using individual choice data, Petrin and Train (2010) and Kim and Petrin (2019) proposed a control function approach, and Berry and Haile (2010) proposed a "special regressor" approach. Incorporating our method into their frameworks to conduct MLE using individual choice data with ξ_{jmt} being embedded is future research.

Therefore, BLP is still a promising method even if consumer-level choice data are available. To reduce the information loss in aggregating individual choice data into market share data, we can divide consumers in a market into different cohorts based on their demographic characteristics and calculate cohort-level product shares (similar to the spirit of Petrin (2002)). The resulting cost is that ξ to be solved by contraction mapping has substantially more dimensions and hence incurs more computation burden. However, the uniqueness and convergence of the solution for ξ are guaranteed.

6. Data

The mortgage data come from Fannie Mae and Freddie Mac. The Fannie Mae and Freddie Mac data cover approximately 12 and 10 million, respectively, single-family first mortgages that were originated between January 2010 and December 2015 and were acquired by Fannie Mae or Freddie Mac. The data have detailed origination information, including the FICO credit score at origination, original debt-to-income ratio (DTI), original unpaid balance, original interest rate, loan term, loan purpose (either for house purchase, cash-out refinance, or non-cash-out refinance), lender, MSA, LTV ratio, etc.

We include loans originated for housing purchases in the sample.[28] We select loans originated in the top 100 MSAs between January 2010 and December 2015. We define a market as an MSA-month combination.

We discretize the LTV ratio into five categories: 0~20%, 20%~40%, 40%~60%, 60%~80%, and higher than 80%. We divide loan terms into three categories: 15 years, 30 years, and all other years. We define a product as a unique combination of lender, LTV category, and loan term category.[29] To reduce the number of products and hence the computational burden of structural

[28] We do not include loans originated for refinance because origination for refinance should be considered as a market different from origination for housing purchases. However, our methodology can be applied to estimate the demand in the refinance market. The market size of the refinance market is the number of all the existing mortgage loans that are eligible for refinance.

[29] LTV is a continuous variable. In the IO literature, researchers discretize continuous characteristics to define products. For example, Berry and Jia (2010) estimated a BLP model for the U.S. airline industry; they discretized airline ticket prices into categories and treated a carrier's tickets on a route within a price category as one product. Moreover, lenders and government-sponsored enterprises (GSEs) usually use a step function of LTV instead of a continuous function of LTV to price mortgages (see the Freddie Mac guarantee fee matrix at https://guide.freddiemac.com/ci/okcsFattach/get/1001717_5 and the Fannie Mae guarantee fee matrix at https://www.fanniemae.com/content/pricing/llpa-matrix.pdf).

estimation, we treat non-fintech lenders other than the largest five lenders (Wells Fargo, Bank of America, Citi, Chase, and U.S. Bank) as one lender; we also treat all fintech lenders as one lender. Accordingly, the total number of products is 105.

The classification of fintech lenders follows Buchak et al. (2018): "We classify a lender as a fintech lender if it has a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender. For example, an applicant to Quicken Loans, the prototypical fintech lender, can be approved for a loan with a locked-in interest rate with no human interaction; the borrower meets a Quicken Loans loan officer for the first time only at closing. An applicant at a non-fintech firm, on the other hand, interacts with a human loan officer much earlier in the process, even if the process begins online. For instance, a borrower may input her name and location online and then be directed to phone a local loan officer to continue." The fintech lenders in the sample include Amerisave Mortgage, Cashcall Inc, Guaranteed Rate Inc, Homeward Residential, Movement Mortgage, and Quicken Loans. There are no "fintech traditional banks."[30]

We collect Moody's Baa corporate bond yield (monthly), yields on 1-year and 10-year treasury constant maturity (monthly), and CPI (monthly) from the Federal Reserve. We use the 1-year treasury constant maturity yield as the baseline interest rate, calculate the credit spread as Moody's Baa corporate bond yield minus 10-year treasury constant maturity yield, and calculate the term spread as 10-year treasury constant maturity yield minus 1-year treasury constant maturity. We deflate borrowers' incomes by CPI and use the deflated incomes as borrowers' demographic characteristics that affect borrowers' price sensitivities (D_i in equation (3.2)). We collect the MSA-level annual employment from the Bureau of Economic Analysis (BEA) and use it as the MSA market size.

The descriptive statistics of important variables are reported in Table 1.

7. Estimation

7.1. First step

In the first step, we estimate how individual-specific mortgage interest rates (prices) are correlated with borrower and loan characteristics. For each month t, we estimate the hedonic pricing equation

[30] Details of fintech lender classification can be found in Online Appendix A.1, A.4, and A.7 in Buchak et al. (2018).

(3.6). The three columns in Table 2 report the regression results using loans originated in January, June, and December, respectively, in 2012. [31] Loans with LTV⩽20%, with loan terms other than 15 or 30 years, or originated by fintech lenders are the omitted reference groups. In general, the mortgage interest rate is increasing in LTV and loan term and is decreasing in the borrower's FICO and income. Conditional on loan and borrower characteristics, fintech lenders charge higher interest rates than non-fintech lenders, which is consistent with the results in Buchak et al. (2018).

The estimates are different across different months due to different macroeconomic conditions such as credit spread and term spread. Based on the estimates, we can predict the borrower-specific price of any product in a borrower's choice set in each month.

7.2. Second step

In the second step, we estimate the structural parameters following the procedure discussed in Sections 3 and 4. We specify the utility function of borrower i in market m in period t for product j as follows:

$$u_{ijmt} = -\alpha_i p_{ijmt} + \lambda_i Fintech_j + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt} \tag{7.1}$$

The borrower-specific price sensitivity α_i is determined as equation (3.3). $Fintech_j$ is the fintech indicator: $Fintech_j$=1 if product j is provided by fintech lenders; $Fintech_j$=0 if product j is provided by non-fintech lenders. The borrower-specific preference for fintech lenders λ_i is determined as follows:

$$\lambda_i = \lambda_0 + v_{1i} \tag{7.2}$$

λ_0 is constant across individual borrowers and is a linear structural parameter. $v_{1i} \sim N(0, \sigma_{v_1}^2)$, where $\sigma_{v_1}^2$ is a nonlinear structural parameter. The existence of v_{1i} allows borrowers to have heterogeneous preferences toward fintech lenders. [32]

[31] Regression results using loans originated in other months are available upon request.

[32] In equation (6.2), we do not allow the borrower-specific preference for fintech lenders λ_i to change according to income. The reason is that, as shown in Buchak et al. (2018), incomes of fintech borrowers are not significantly different from incomes of non-fintech borrowers. Moreover, fintech borrowers have very similar default rates as non-fintech borrowers (see Buchak et al., 2018).

X_j in equation (7.1) includes the intercept, the LTV category indicators,[33] the loan term category indicators,[34] and the dummy variables for each of the largest five non-fintech lenders (Wells Fargo, Bank of America, Citi, Chase, and U.S. Bank).[35] The total number of products is 105.[36]

We employ the following variables as the instruments for price: baseline interest rate, credit spread, term spread, the interaction between credit spread and the LTV indicators, the interaction between term spread and the loan term indicators, and the average interest rate of mortgages in the same LTV and maturity category originated by the same lender in other cities during the month.

The estimation results based on the modified BLP with price bias correction (discussed in Section 4), the one without price bias correction (discussed in Section 3), and the traditional BLP, respectively, are reported in Columns 1, 2, and 3 of Table 3.[37] The estimates of γ (the effect of income in reducing borrowers' price sensitivities) based on these three methods are 15.7006, 16.7458, and 17.4588, respectively.

The estimated γ based on the modified BLP without price bias correction (16.7458) is smaller than that based on the traditional BLP (17.4588). This pattern is consistent with the discussion in Section 3.3.

The estimated γ based on the modified BLP with price bias correction (15.7006) is smaller than that based on the modified BLP without price bias correction (16.7458). The reason is that as shown in equation (4.10), the bias-correction component of the predicted price has the term $-\gamma\sigma_{\eta_t}^2 D_i$; if we ignore it, its effect will be captured by the term $-\gamma D_i$ in α_i, leading to a greater estimate of γ.

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[33] The dummy variables for each of the following LTV categories: 20%~40%, 40%~60%, 60%~80%, and higher than 80%, with 0~20% as the omitted reference group.

[34] The dummy variables for 15-year loans and 30-year loans, with loans with other terms as the omitted reference group.

[35] Non-fintech lenders other than the largest five is the omitted reference group. As we already have a constant λ_0 embedded before $Fintech_j$ in equation (7.1), we do not need to include the dummy variable for fintech lenders in X_j.

[36] One may argue that borrowers may not search all the products available in the market and hence the choice set in consideration is smaller than the set of available products. However, many mortgages are originated through mortgage brokers. Mortgage brokers know the information of most products and can help choose one that best fits a borrower's need.

[37] For the traditional BLP, we use the average interest rates of all the loans within a product category in a market as the identical price faced by all the borrowers in the market for the product.

To summarize, the estimate of γ generated by the traditional BLP is approximately 11.20% greater than that generated by the modified BLP with price bias correction and is approximately 4.26% greater than that generated by the modified BLP without price bias correction.

Figure 1 displays the relationships between individual-specific price sensitivity (α_i) and income separately based on the estimates of the modified BLP with price bias correction, the modified BLP without price bias correction, and the traditional BLP, respectively. Each point represents a combination of α_i and income for a borrower in the sample. Compared with our methods, the traditional BLP method tends to overestimate price sensitivities of low-income borrowers and underestimate price sensitivities of high-income borrowers.

The three lines in Figure 1 become thinner as income increases because there are more low-income borrowers than high-income borrowers in the population (see Figure C.1 in Appendix C for the distribution of income). Therefore, as displayed in Figure 2, compared with the distributions of individual-specific price sensitivity (α_i) recovered by our methods, the distribution recovered by the traditional BLP method has more borrowers with high price sensitivity and fewer borrowers with low price sensitivity.[38]

The estimates of σ_v based on the modified BLP with price bias correction, the modified BLP without price bias correction, and the traditional BLP, respectively, are 3.3332, 2.1063, and 2.8681. The reason why the third method generates a larger σ_v than the second method is that if we ignore the across-individual price dispersion within a product, the heterogeneity in price sensitivities will be enlarged to capture the ignored price dispersion. The reason why the second method generates a smaller σ_v than the first method is that if we do not conduct price bias correction, the predicted prices tend to be in a lower domain in which the derivative of demand with respect to price tends

[38] In this empirical setting, the results turn out that the blue (or orange) line and the yellow line in Figure 1 intersect approximately in the middle. Theoretically, the two lines can intersect close to the lower or upper bound of the income domain. If the two lines intersect close to the upper bound of the income domain, the distribution of individual-specific price sensitivity (α_i) recovered by the traditional BLP method could have more borrowers with high price sensitivity and fewer borrowers with low price sensitivity than the distribution recovered by our methods. In contrast, if the two lines intersect close to the lower bound of the income domain, the distribution recovered by the traditional BLP method could have fewer borrowers with high price sensitivity and more borrowers with low price sensitivity than the distribution recovered by our methods.

to be larger in magnitude;[39] to fit the derivative reflected by the actual data, on the one hand, the overall levels of price sensitivities α_i for most individuals ($\alpha_0 - \gamma D_i$) should move upward (compare the blue line with the orange line in Figure 1); on the other hand, the unobserved heterogeneity in price sensitivities σ_v should be reduced such that there are fewer individuals with extremely low v_i.

The estimates of other parameters are consistent with theory or intuition. Borrowers obtain higher utility from products with longer loan terms or higher LTVs because these products will allow borrowers to have lower amounts of down payments or monthly debt repayments.

The estimates tend to be large in magnitude. The first reason is that the price variable (interest rate) is small in magnitude (at the level of 10^{-2}). The second reason is that the market share of each product is small, given that the market size is defined as the MSA-level employment.

The standard errors are very small in magnitude relative to the estimates. The reason is that the number of products marketed is large: each MSA in each month is considered as one market and the sample covers 72 months and 100 MSAs. The magnitudes of standard errors mostly come from V_1 in equation (3.15). V_2 and V_3 are much smaller in magnitude than V_1.

8. An application on quantifying borrowers' willingness to pay for fintech

In this section, we conduct one of the possible further applications of our methodology: quantifying borrowers' willingness to pay for fintech based on the estimates of the structural parameters in Section 7.[40]

8.1. Background

After 2010, the U.S. residential mortgage industry experienced a wave of technological innovation (e.g., automation, big data, and artificial intelligence). Multiple fintech lenders were established and have become an increasingly important source of mortgage credit to U.S. households. Unlike traditional mortgage lenders that heavily rely on the network of "bricks and mortar" branches or

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[39] To see this simply, $\frac{d\left[\frac{\exp\{-\alpha_i p\}}{1+\exp\{-\alpha_i p\}}\right]}{dp} = \frac{-\alpha_i \exp\{-\alpha_i p\}}{(1+\exp\{-\alpha_i p\})^2} < 0$ and $\frac{d^2\left[\frac{\exp\{-\alpha_i p\}}{1+\exp\{-\alpha_i p\}}\right]}{dp^2} = \frac{{\alpha_i}^2 \exp\{-\alpha_i p\}(1-\exp\{-\alpha_i p\})}{(1+\exp\{-\alpha_i p\})^3} > 0$ for $p > 0$ and $\alpha_i > 0$.

[40] Previous studies have applied the traditional BLP framework to quantifying automobile purchasers' willingness to pay for domestic brands (home bias). See Barwick et al. (2021), Cosar et al. (2018), and Goldberg and Verboven (2001).

local brokers, fintech lenders have a centralized underwriting system that can accomplish a complete end-to-end online mortgage application and approval process. They can electronically collect documentation about borrowers' income, assets, and credit history and then make approval decisions in minutes. Fuster et al. (2019) found that fintech lenders process mortgage applications 20% faster than non-fintech lenders.

According to Fuster et al. (2019), the market share of fintech mortgage lenders has grown from 2% ($34 billion) to 8% ($161 billion) from 2010 to 2016; as of the end of 2016, all fintech lenders were independent mortgage originators that securitize mortgages in the primary market and operated without relying on deposit financing or branch networks. Using loan-level data, Buchak et al. (2018) found that fintech lenders charge higher interest rates than non-fintech lenders, controlling for observable characteristics. Fuster et al. (2019) found no evidence that fintech lenders target borrowers with low access to finance.

8.2 Quantifying borrowers' willingness to pay for fintech

In this subsection, we quantify borrowers' willingness to pay for fintech based on the estimates of the structural parameters in Section 7.

According to equation (8.1), for borrower i, given choosing mortgage products in the same LTV-maturity category, the difference between the utility of choosing the product of fintech lenders (product j') and the utility of choosing a product of a certain non-fintech lender h among the largest five (product j) is

$$\begin{aligned} u_{ij'mt} - u_{ijmt} = -\alpha_i p_{ij'mt} + \lambda_i + \xi_{j'mt} + \varepsilon_{ij'mt} - (-\alpha_i p_{ijmt} + lender_h \\ + \xi_{jmt} + \varepsilon_{ijmt}) \end{aligned} \tag{8.1}$$

where $lender_h$ is the lender fixed effect for lender h. Making the utility difference equal to zero, we can solve the premium that borrower i is willing to pay for choosing fintech lenders versus non-fintech lender h for originating a mortgage in this LTV-maturity category as follows:

$$premium_{ijj'mt} = \frac{\lambda_i + \xi_{j'mt} + \varepsilon_{ij'mt} - (lender_h + \xi_{jmt} + \varepsilon_{ijmt})}{\alpha_i} \tag{8.2}$$

For the category of 30-year mortgages with LTV within 60%~80% (the most popular category),[41] we calculate $premium_{ijj'mt}$ across all borrowers in all markets. The five panels in Figure 3 display the distributions of borrowers' willingness to pay for choosing fintech lenders versus each of the five largest non-fintech lenders, respectively. The majority of borrowers still have a negative willingness to pay for fintech lenders versus a non-fintech lender, which is consistent with the fact that the market share of fintech lenders was below 10% in the sample period.

The results indicate that the positive premium charged by fintech lenders documented in the literature (Buchak et al., 2018) should possibly be explained by that fintech lenders exploit borrowers who have strong preferences for application convenience and speed and hence have a positive willingness to pay for fintech. The distribution can help fintech lenders develop pricing strategies and help traditional non-fintech lenders evaluate the competition they face from fintech lenders.

The distribution of willingness to pay for fintech is determined by multiple factors, including the estimates of λ_0, $lender_h$, and σ_{v_1} and the distributions of α_i, ξ_{jmt}, and $\xi_{j'mt}$. As shown in Figure 3, compared with the distributions recovered by our methods, the distribution recovered by the traditional BLP method tends to have more borrowers with low willingness to pay and fewer borrowers with high willingness to pay.

Figure 4 compares Wells Fargo with the other four major non-fintech lenders for the results generated by the modified BLP with price bias correction. The distribution of borrowers' willingness to pay for choosing fintech lenders versus Wells Fargo tends to have more borrowers in the lower willingness part (the left part) than that versus any of the other four major non-fintech lenders. The reason is that Wells Fargo has the largest lender fixed effect in borrowers' utility function (see the estimate in Table 3), which is consistent with Wells Fargo's largest market share (see the descriptive statistics in Table 1).

8.3 Value of face-to-face interaction for first-time borrowers

[41] We choose this mortgage category to display because it is the most popular category. According to Table 1, the market share of mortgages with LTV within 60%~80% is above 50%; the market share of 30-year mortgages is approximately 90%.

Buchak et al. (2018) and Fuster et al. (2019) found reduced-form evidence that first-time homebuyers are less likely to choose fintech lenders. The reason is that without experience of mortgage origination procedures (lack of financial literacy), first-time borrowers usually prefer to interact face-to-face with lenders, rather than applying online. In this subsection, by structural estimation, we quantify the value of face-to-face interaction for a first-time borrower to borrow from a non-fintech lender.

We add additional components to equation (7.1) and it becomes the following equation:

$$u_{ijmt} = -\alpha_i p_{ijmt} + (\lambda_0 + v_{1i})Fintech_j + \lambda_0' \cdot 1stTimeHomebuyer_i \cdot \left(1 - Fintech_j\right) + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt} \tag{8.3}$$

where $1stTimeHomebuyer_i = 1$ if borrower i is a first-time homebuyer and $1stTimeHomebuyer_i = 0$ otherwise. The parameter λ_0' measures the additional utility that a first-time borrower obtains from the convenience or financial-literacy gains from face-to-face interaction with non-fintech lenders relative to borrowing online.[42]

In the first-step estimation of the hedonic pricing equation (3.6), we add additional control variables, including the first-time-homebuyer status and property occupancy status (primary residence, second home, or investment property). The estimation results are reported in Table C.1 in Appendix C. The coefficients are consistent with intuition. The interest rate of mortgages for investment property is higher than that for second home, which is higher than that for primary residence. The reason is that the incentive to default strategically decreases from the first one to the third one. With the property occupancy status being controlled for, first-time homebuyers are charged higher interest rates because of their high likelihood of having illiquidity-triggered defaults.

Table 4 reports the second-step estimation results. The estimated $\lambda_0{}'$ based on the traditional BLP (8.5647) is greater than that based on the modified BLP either with or without price bias correction (5.1099 or 3.8112). The reason is that, on average, first-time homebuyers face lower prices than non-first-time homebuyers for the same products, given that first-time homebuyers' mortgage collaterals are their primary residence and hence their incentives to default strategically

[42] Equation (8.3) can be rewritten as

$$u_{ijmt} = -\alpha_i p_{ijmt} + (\lambda_0 - \lambda_0' \cdot 1stTimeHomebuyer_i + v_{1i})Fintech_j + \lambda_0' \cdot 1stTimeHomebuyer_i + \beta X_j + \xi_{jmt} + \varepsilon_{ijmt}$$

Therefore, equation (8.3) is equivalent to allowing the random coefficient of the fintech lender indicator (λ_i in equation (7.1)) to depend on the first-time homebuyer status of borrower i.

are smaller;[43] the traditional BLP ignores the lower prices faced by first-time homebuyers and hence overestimates the disutility of paying prices; consequently, the utility of face-to-face interaction will be overestimated to compensate.

The estimated $\lambda_0{}'$ based on the modified BLP without price bias correction (3.8112) is smaller than that based on the modified BLP with price bias correction (5.1099). The reason is that the downward price bias, $-\alpha_i\sigma_{\eta_t}^2$, is greater in magnitude for first-time homebuyers, given that they are usually low-income borrowers and are more price-sensitive (larger α_i). Without correcting the bias, we tend to underestimate their disutility of paying prices to a larger extent than we do for non-first-time homebuyers'; thus, their utility of face-to-face interaction will be underestimated to compensate.

Figure 5 displays the distribution of the value (in terms of bps of interest rates) of face-to-face interaction when a first-time borrower borrows from a non-fintech lender, i.e., the distribution of $\frac{\lambda_0{}'}{\alpha_i}$. As shown by the blue histogram (based on the modified BLP with price bias correction), for most first-time borrowers, the utility of face-to-face interaction is equivalent to that of paying from 125 to 200 bps lower interest rates. This result generates a managerial implication that it is substantially worthwhile for fintech lenders to consider posting videos on their websites showing mortgage borrowing steps in detail and addressing possible concerns to increase their market shares over first-time homebuyers. Fintech lenders may also consider investing in the provision of online video chat consultation services.

The estimates of γ in Table 4 maintain the same pattern as those in Table 3: the estimated γ based on the modified BLP with price bias correction is smaller than that based on the modified BLP without price bias correction, which is smaller than that based on the traditional BLP.

Comparisons between the structural parameters in Tables 3 and 4 make sense. σ_{v_1} in Table 4 is smaller than that in Table 3. The reason is that as equation (8.3) is equivalent to allowing borrowers' random preference toward fintech lenders to depend on an additional observable ($1stTimeHomebuyer_i$), the unobserved heterogeneity in borrowers' preference for fintech

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[43] As shown in Table C.1 in Appendix C for the regression of the hedonic pricing equation, when we control for the property occupancy status, the first-time homebuyer indicator is significantly positive because first-time homebuyers usually have a lower net worth and thereby are more likely to have illiquidity-triggered defaults. However, as shown in Table C.2 in Appendix C, if we do not control for the property occupancy status, the first-time homebuyer indicator becomes significantly negative (see Ma (2016) for more discussion).

lenders becomes smaller. σ_v in Table 4 is larger than that in Table 3. The reason is that as we add additional regressors (the first-time homebuyer status and the occupancy status) in the first-step estimation for the hedonic pricing equation, the fit of the equation is improved and the root mean squared error $\hat{\sigma}_{\eta_t}$ becomes smaller; thus, the unobserved heterogeneity in price sensitivities needs to increase to compensate for the reduction in the unobserved price dispersion conditional on observable demographic characteristics.

9. Discussion

In this section, we discuss several other potential further applications of the methodology developed in this paper.

9.1. Mergers

Welfare analyses on mergers are important issues in IO research and antitrust regulation. Theoretically, mergers can generate two competing effects on consumer surplus. First, mergers can improve firms' operating efficiency, which tends to increase consumer surplus. Second, mergers can increase firms' market power, which tends to decrease consumer surplus. Counterfactual analyses based on structural model estimates help antitrust regulators make decisions on whether to approve a merger proposal.

For financial products, as the traditional BLP model overestimates price sensitivities of low-income borrowers and underestimates price sensitivities of high-income borrowers, it will overestimate the welfare loss for low-income borrowers and underestimate the welfare loss for high-income borrowers caused by price increases after a merger. Whether the total welfare loss for all borrowers in a market is overestimated or underestimated depends on the income distribution of the borrowers in the market.

9.2. Deregulation and market entry

Bank branch deregulation in the U.S. from 1990 to 2010 allowed more entries into local markets, which made local markets more competitive. Multiple studies have provided reduced-form evidence that bank branch deregulation in the U.S. reduced mortgage interest rates (Sun and Yannelis, 2016), increased mortgage accessibility and homeownership (Tewari, 2014; Favara and Imbs, 2015; Hacamo, 2021), increased college loan take-ups and college enrollments (Sun and

Yannelis, 2016), and reduced interest rates of small-business loans (Rice and Strahan, 2010). However, little research has estimated structural models and conducted welfare analyses.

The methodology developed in this paper can enable researchers to conduct structural estimation and welfare analyses for banking deregulation. The advantage of this method over the traditional BLP is that it will not overestimate the welfare increase for low-income borrowers and will not underestimate the welfare increase for high-income borrowers caused by price drops after deregulation.

The procedures of welfare analyses based on estimates of our modified BLP model can follow those based on estimates of traditional BLP models in the literature, such as the welfare analyses conducted by Nevo (2000a) for mergers and by Petrin (2002) for product entries.

9.3. Financial inclusion

Low-income (financially disadvantaged) populations usually have limited financial access despite strong needs. On the one hand, low-income residents are of high risk and hence lenders would charge high interest rates for loans originated to them. On the other hand, low-income residents are price-sensitive and cannot afford loans with high interest rates. Therefore, the market itself usually cannot come up with a financial product that is both affordable to low-income borrowers and profitable to lenders.

Given that whether the benefits produced by financial intermediation and markets can spread widely throughout the population is a crucial concern of governments, multiple programs are initiated by governments to enhance low-income residents' access to affordable credits. For example, Mortgage Revenue Bonds (MRBs) are tax-exempt bonds that state and local governments issue through housing finance agencies (HFAs) to help fund below-market-interest-rate mortgages for low-to-moderate-income first-time homebuyers (Ergungor, 2010). The U.S. Department of Education's federal student loan program makes loans with interest rates lower than the market rates to eligible students to cover their educational expenses. The Canada Small Business Financing Program improves small business owners' access to loans from financial institutions by sharing the risk with lenders.

The methodology developed in this paper can enable researchers to conduct welfare analyses for these programs. It can help evaluate welfare gains of low-income borrowers, program expenses borne by governments, and welfare changes of high-income borrowers due to changes in lenders' pricing strategies in response to changes in aggregate demand caused by these programs.

9.4. Price discrimination in nonfinancial-product markets

Our methodology can be applied not only to estimate the demand for differentiated financial products but also to estimate the demand for differentiated nonfinancial products with price discrimination based on consumers' demographic characteristics.

Price discrimination has been documented in many nonfinancial-product markets.[44] If firms conduct price discrimination based on consumers' demographic characteristics, the underlying assumption of the traditional BLP model that all consumers in a market face the same price for the same product will not hold. In this situation, a consumer's demographic characteristics affect both the consumer's price sensitivity and the individual-specific prices the consumer faces, very similar to the situation of financial-product markets. Then, our methodology can be applied to estimate the demand. The data needed for estimation include individual-level price data for a random sample of consumers and the demographic characteristics of these consumers; data on characteristics and market shares of each product; and data on the distribution of demographic characteristics of the population in each market.

Unlike D'Haultfœuille et al. (2019), our method leverages observed individual-specific prices rather than recovers them from a contraction mapping; consequently, our method does not need the assumption that consumers in a subgroup with the same demographic characteristics always obtain identical price offers from any seller. Prices in our model are heterogeneous at the individual level instead of at the subgroup level. Therefore, our model can accommodate cases such as that among consumers with the same demographic characteristics, some visit stores on sale days whereas others visit on non-sale days, which is common in the real world. Because our method leverages observed individual-specific prices, our method needs to handle the selectivity issue that individual-specific prices of only products chosen by individuals are observed by econometricians.

[44] For example, Borenstein and Rose (1994), Gerardi and Shapiro (2009), and Kim et al. (2022) on the airline industry; Busse and Rysman (2005) on the Yellow Pages advertising industry; Delgado and Waterson (2003) on the retail outlets industry; Borenstein (1991) and Lewis (2008) on the retail gasoline industry; Nevo and Wolfram (2002) on the cereals industry; Leslie (2002) on the Broadway theater industry; and Asplund, Eriksson and Strand (2008) on the newspaper industry

10. Conclusion

The well-known BLP model has been used to estimate demand for differentiated products in many nonfinancial industries. However, little research has applied the BLP framework to financial-product industries, although many financial-product industries have differentiated products. One underlying assumption in the BLP model is that all consumers in a market face the same price for the same product, which is natural for nonfinancial-product industries (e.g., the automobile and grocery industries). However, in financial-product industries, different consumers usually face different prices for the same financial product, and the price a consumer faces is related to the consumer's demographic characteristics (individual-risk-based pricing). Meanwhile, the consumer's price elasticity of demand is also related to its demographic characteristics. For example, mortgage borrowers with lower incomes tend to be charged higher prices (interest rates) by lenders because of higher default risk; meanwhile, people with lower incomes are more price-sensitive or more price-elastic.

If we simply apply the BLP model to estimate financial product demand and assume that borrowers face the same price for the same loan product, we will overestimate the effect of income in reducing borrowers' price sensitivities. In other words, we will overestimate price sensitivities of low-income borrowers and underestimate price sensitivities of high-income borrowers. Borrowers with higher incomes are more likely to purchase a financial product for two reasons: first, they have lower price sensitivities; second, they face lower prices because of their lower default risk. If we ignore the channel that customers with higher incomes face lower prices, we have to make these customers even less price-sensitive in the model to fit the data.

In this paper, we propose a two-step estimation for demand in financial-product markets in which different consumers face different prices for the same financial product and the price a consumer faces is related to the consumer's demographic characteristics. In addition, we propose a method to correct biases arising from the selectivity that only the individual-specific price of the product chosen by an individual is observed by econometricians; individual-specific prices of products not chosen by the individual cannot be observed by econometricians while can be observed by the individual when she/he compares among products.

We apply our method to demand estimation of the U.S. residential mortgage market. The estimate of the income effect on price sensitivities generated by the traditional BLP is 11.20% greater in magnitude than that generated by our method.

Given that there are more low-income borrowers than high-income borrowers in the population, the distribution of individual-specific price sensitivities recovered by the traditional BLP method has more high-price-sensitivity borrowers and fewer low-price-sensitivity borrowers than the distribution recovered by our method.

The methodology developed in this paper enables researchers to extend the BLP framework into financial-product industries and analyze typical issues in modern IO for these industries, such as welfare analyses on bank mergers, bank branch deregulations, and governmental programs aiming to enhance financial access of low-income (financially disadvantaged) populations. The methodology can also be applied to demand estimation for nonfinancial-product markets with price discrimination based on consumers' demographic characteristics.

We conduct one of the possible further applications of our methodology: quantifying borrowers' willingness to pay for fintech. Based on the structural estimates, we recover the distributions of borrowers' willingness to pay for fintech. The results indicate that the positive premium charged by fintech lenders documented in the literature should possibly be explained by that fintech lenders exploit the small proportion of borrowers who have strong preferences for application convenience and speed and hence have a positive willingness to pay for fintech. The results can help fintech lenders develop pricing strategies and help traditional non-fintech lenders evaluate the competition they face from fintech lenders.

We further use our structural estimation framework to quantify the value of face-to-face interaction for a first-time borrower to borrow from a non-fintech lender. The utility of the convenience or financial-literacy gains from face-to-face interaction for first-time borrowers is equivalent to that of paying from 125 to 200 bps lower interest rates. This result generates a managerial implication that it is substantially worthwhile for fintech lenders to consider posting videos on their websites showing mortgage borrowing steps in detail and addressing possible concerns to increase their market shares over first-time homebuyers. Fintech lenders may also consider investing in the provision of online video chat consultation services.

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Table 1. Descriptive statistics

	N	Mean	Std Dev
Panel A: Product characteristics			
Interest rate (price)	5324732	0.0424	0.0055
LTV			
$0 < LTV \leq 20\%$	5324732	0.0021	0.0021
$20\% < LTV \leq 40\%$	5324732	0.0229	0.0224
$40\% < LTV \leq 60\%$	5324732	0.0758	0.0701
$60\% < LTV \leq 80\%$	5324732	0.5693	0.2452
$LTV > 80\%$	5324732	0.3299	0.2211
Loan term			
15 years	5324732	0.0903	0.0821
30 years	5324732	0.8922	0.0962
Others	5324732	0.0175	0.0172
Lender			
Bank of America	5324732	0.0397	0.0381
Citi	5324732	0.0459	0.0438
Chase	5324732	0.0439	0.0420
U.S. Bank	5324732	0.0388	0.0373
Wells Fargo	5324732	0.2167	0.1697
Other non-fintech lenders	5324732	0.5970	0.2406
Fintech lenders	5324732	0.0180	0.0177
Panel B: Borrower demographics			
Monthly income	5324732	3921.16	3044.57
Deflated monthly income (dollars in January 2010)	5324732	3691.26	2887.13
FICO	5324732	759.86	59.06
1st-time homebuyer	5324732	0.3608	0.4802
Primary residence	5324732	0.8446	0.3623
Second home	5324732	0.0545	0.2270
Investment property	5324732	0.1009	0.3012
Panel C: Macroeconomic variables			
Baseline interest rate	72	0.0021	0.0011
Credit spread	72	0.0279	0.0033
Term spread	72	0.0226	0.0056

The descriptive statistics in panels A and B are based on single-family first mortgages that were originated in the top 100 MSAs between January 2010 and December 2015 for housing purchases and were acquired by Fannie Mae or Freddie Mac. The descriptive statistics in panel C are based on monthly national macroeconomic variables from January 2010 to December 2015.

Table 2. The first-step estimation: hedonic pricing equations

Dependent variable: interest rate	Column 1 January, 2012	Column 2 June, 2012	Column 3 December, 2012
LTV			
$20\% < LTV \leq 40\%$	-0.0001***	-0.0003***	-0.0003***
	(0.0004)	(0.0002)	(0.0002)
$40\% < LTV \leq 60\%$	0.0000***	-0.0003***	-0.0003***
	(0.0004)	(0.0002)	(0.0002)
$60\% < LTV \leq 80\%$	0.0016***	0.0010***	0.0010***
	(0.0004)	(0.0002)	(0.0002)
$LTV > 80\%$	0.0012***	0.0009***	0.0008***
	(0.0004)	(0.0002)	(0.0002)
Loan term			
15 years	-0.0035***	-0.0038***	-0.0042***
	(0.0002)	(0.0001	(0.0001)
30 years	0.0030***	0.0034***	0.0023***
	(0.0001)	(0.0001	(0.0001)
Lender			
Bank of America	-0.0017***	-0.0015***	-0.0009***
	0.0003)	(0.0001)	(0.0001)
Citi	-0.0012***	-0.0006***	-0.0013***
	(0.0002)	(0.0001)	(0.0001)
Chase	-0.0019***	-0.0012***	-0.0007***
	(0.0003)	(0.0001)	(0.0001)
U.S. Bank	-0.0022***	-0.0016***	-0.0015***
	(0.0002)	(0.0001)	(0.0001)
Wells Fargo	-0.0013***	-0.0013***	-0.0015***
	(0.0002)	(0.0001)	(0.0001)
Other non-fintech lenders	-0.0016***	-0.0010***	-0.0010***
	(0.0002)	(0.0001)	(0.0001)
FICO/1000	-0.0085*	-0.0166***	-0.0187***
	(0.0049)	(0.0003)	(0.0003)
Monthly income ($ 1,000)	-0.0001***	-0.0001***	-0.0001***
	(0.0000)	(0.0000)	(0.0000)
MSA fixed effects	✓	✓	✓
Root mean squared error $\hat{\sigma}_{\eta_t}$	0.0033	0.0027	0.0027
R squared	0.3506	0.4576	0.4220
N	39817	85590	69330

For each month t, we estimate the hedonic pricing equation (3.6). The regressors include product characteristics and borrower characteristics. Columns 1, 2, and 3 in this table report the regression results using loans originated in January, June, and December, respectively, in 2012. Regression results using loans originated in other months are available upon request. Standard errors are reported in parentheses. * denotes significance at a 10% level. ** denotes significance at a 5% level. *** denotes significance at a 1% level. Loans with $LTV \leq 20\%$, with loan terms other than 15 or 30 years, or originated by fintech lenders are the omitted reference groups.

Table 3. The second-step estimation: structural parameters for mortgage markets

	Column 1 Modified BLP with price bias correction	Column 2 Modified BLP without price bias correction	Column 3 Traditional BLP
Nonlinear structural parameters			
α_0	389.8813***	409.7835***	417.1150***
	(1.5290)	(4.4591)	(1.8689)
γ	15.7006***	16.7458***	17.4588***
	(0.0999)	(0.0624)	(0.0742)
σ_v	3.3332***	2.1063***	2.8681***
	(0.0323)	(0.0152)	(0.0165)
σ_{v_1}	4.1652***	4.1096***	5.7129***
	(0.0229)	(0.0338)	(0.0257)
Linear structural parameters			
LTV			
$20\% < LTV \leq 40\%$	82.8553***	84.5181***	87.1252***
	(2.0171)	(3.9017)	(1.6012)
$40\% < LTV \leq 60\%$	144.2478***	147.1609***	151.6927***
	(3.5092)	(6.1313)	(4.2049)
$60\% < LTV \leq 80\%$	230.5267***	235.254***	242.8124***
	(5.9891)	(10.9467)	(4.4544)
$LTV > 80\%$	189.9385***	193.8487***	200.0927***
	(5.9791)	(7.7175)	(3.7575)
Loan term			
15 years	32.3202***	32.8728***	33.3703***
	(1.0845)	(1.6246)	(0.8176)
30 years	125.1298***	127.6262***	132.0924***
	(3.8494)	(5.5739)	(3.2807)
Lender fixed effects			
Bank of America	-385.636***	-393.7581***	-407.6354***
	(10.7743)	(21.3303)	(8.9089)
Citi	-130.637***	-131.0838***	-134.9265***
	(6.6893)	(11.1478)	(6.1795)
Chase	-794.071***	-809.9947***	-835.9345***
	(19.0278)	(31.2138)	(17.8178)
U.S. Bank	-880.485***	-900.2326***	-927.6021***
	(21.3733)	(41.7021)	(22.0208)
Wells Fargo	189.441***	193.9538***	201.3432***
	(6.3218)	(11.7227)	(3.7377)
Fintech lenders (λ_0)	-703.467***	-718.7226***	-742.7221***
	(18.6126)	(26.7547)	(13.3646)
Intercept	2.1687***	2.1458***	1.9695***
	(0.0392)	(0.0603)	(0.0355)

This table reports the estimates of structural parameters in borrowers' utility function as defined in equations (7.1), (7.2), and (3.3), including both nonlinear structural parameters $\{\alpha_0, \gamma, \sigma_v, \sigma_{v_1}\}$ and linear structural parameters β and λ_0. Standard errors are reported in parentheses. * denotes significance at a 10% level. ** denotes significance at a 5% level. *** denotes significance at a 1% level. Loans with $LTV \leq 20\%$, with loan terms other than 15 or 30 years, or originated by non-fintech lenders other than the largest five lenders are the omitted reference groups. Column 1 reports the estimation results based on the modified BLP with correction for the price prediction biases arising from the selectivity that only the individual-specific price of the product chosen by an individual is observed and those of products not chosen by the individual cannot be observed by econometricians (discussed in Section 4). Column 2 reports the results based on the modified BLP without price bias correction (discussed in Section 3). Column 3 reports the results based on the traditional BLP; we use the average interest rates of all the loans within a product category in a market as the identical price faced by all the borrowers in the market for the product for the traditional BLP estimation. The estimate of γ (the effect of income on price sensitivities) generated by the traditional BLP is approximately 11.20% greater than that generated by the modified BLP with price bias correction and is approximately 4.26% greater than that generated by the modified BLP without price bias correction.

Table 4. Quantifying the value of face-to-face interaction for first-time borrowers: the second-step estimation

	Column 1 Modified BLP with price bias correction	Column 2 Modified BLP without price bias correction	Column 3 Traditional BLP
Nonlinear structural parameters			
α_0	389.4422***	418.4240***	432.8320***
	(1.0062)	(3.9669)	(2.6344)
γ	15.2994***	17.3647***	18.2164***
	(0.0738)	(0.0708)	(0.1909)
σ_v	25.6731***	2.5339	25.1447***
	(0.2103)	(5.8974)	(0.3985)
σ_{v_1}	1.1492***	2.4867***	1.0208***
	(0.0125)	(0.0178)	(0.0071)
$1stTimeHomebuyer_i \cdot (1 - Fintech_j)$	5.1099***	3.8112***	8.5647***
	(0.0239)	(0.0400)	(0.1011)
Linear structural parameters			
LTV			
$20\% < LTV \leq 40\%$	82.7513***	83.8796***	87.8536***
	(1.3828)	(3.2325)	(2.2053)
$40\% < LTV \leq 60\%$	144.0547***	146.0330***	152.9292***
	(2.2420)	(6.3344)	(5.6553)
$60\% < LTV \leq 80\%$	230.1634***	233.4060***	244.6747***
	(3.4524)	(10.8792)	(5.7603)
$LTV > 80\%$	189.6383***	192.3223***	201.6179***
	(2.9371)	(7.8771)	(4.8198)
Loan term			
15 years	32.3407***	32.6738***	33.8073***
	(0.4800)	(1.7940)	(1.0414)
30 years	124.9656***	126.6509***	133.1058***
	(1.6316)	(5.5743)	(3.5460)
Lender fixed effects			
Bank of America	-384.8882***	-390.6792***	-410.7920***
	(6.1960)	(15.8789)	(12.4915)
Citi	-132.0887***	-131.4033***	-138.9814***
	(3.4050)	(8.5580)	(7.3083)

Chase	-792.9743***	-803.5855***	-841.9338***
	(10.3573)	(31.7579)	(18.7194)
U.S. Bank	-878.0050***	-892.2731***	-933.0791***
	(13.0814)	(40.7940)	(19.9376)
Wells Fargo	188.7474***	192.0907***	202.1126***
	(2.6292)	(9.8239)	(4.2198)
Fintech lenders (λ_0)	-701.7421***	-712.6292***	-747.3383***
	(8.5044)	(31.5086)	(25.6310)
Intercept	2.2188***	2.1747	2.1047***
	(0.0402)	(5.9010)	(0.0719)

This table reports the structural estimates for equation (8.3). $1stTimeHomebuyer_i = 1$ if borrower i is a first-time homebuyer and $1stTimeHomebuyer_i = 0$ otherwise. The coefficient of $1stTimeHomebuyer_i \cdot (1 - Fintech_j)$, λ_0', measures the additional utility that a first-time borrower obtains from the convenience or financial-literacy gains from face-to-face interaction with non-fintech lenders relative to borrowing online. Standard errors are reported in parentheses. * denotes significance at a 10% level. ** denotes significance at a 5% level. *** denotes significance at a 1% level.

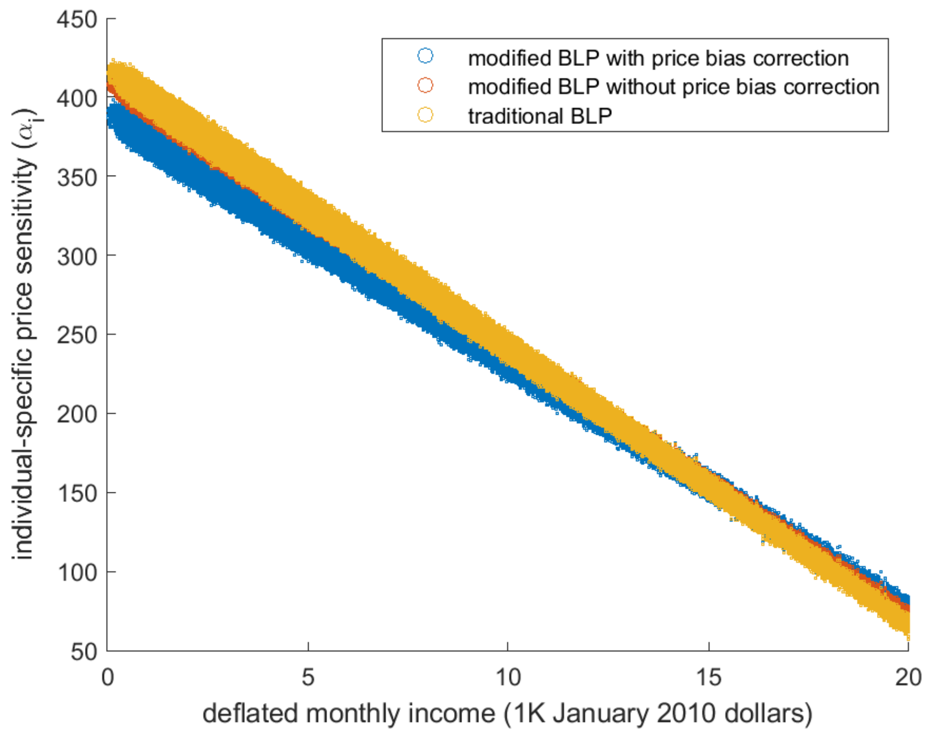

Figure 1. Relationship between individual-specific price sensitivity (α_i) and income (please see the color figure). Each point represents a combination of α_i and income for a borrower in the sample. The results generated by the modified BLP with price bias correction, the one without price bias correction, and the traditional BLP, respectively, are displayed as blue, orange, and yellow points. Compared with our methods, the traditional BLP method tends to overestimate price sensitivities of low-income borrowers and underestimate price sensitivities of high-income borrowers.

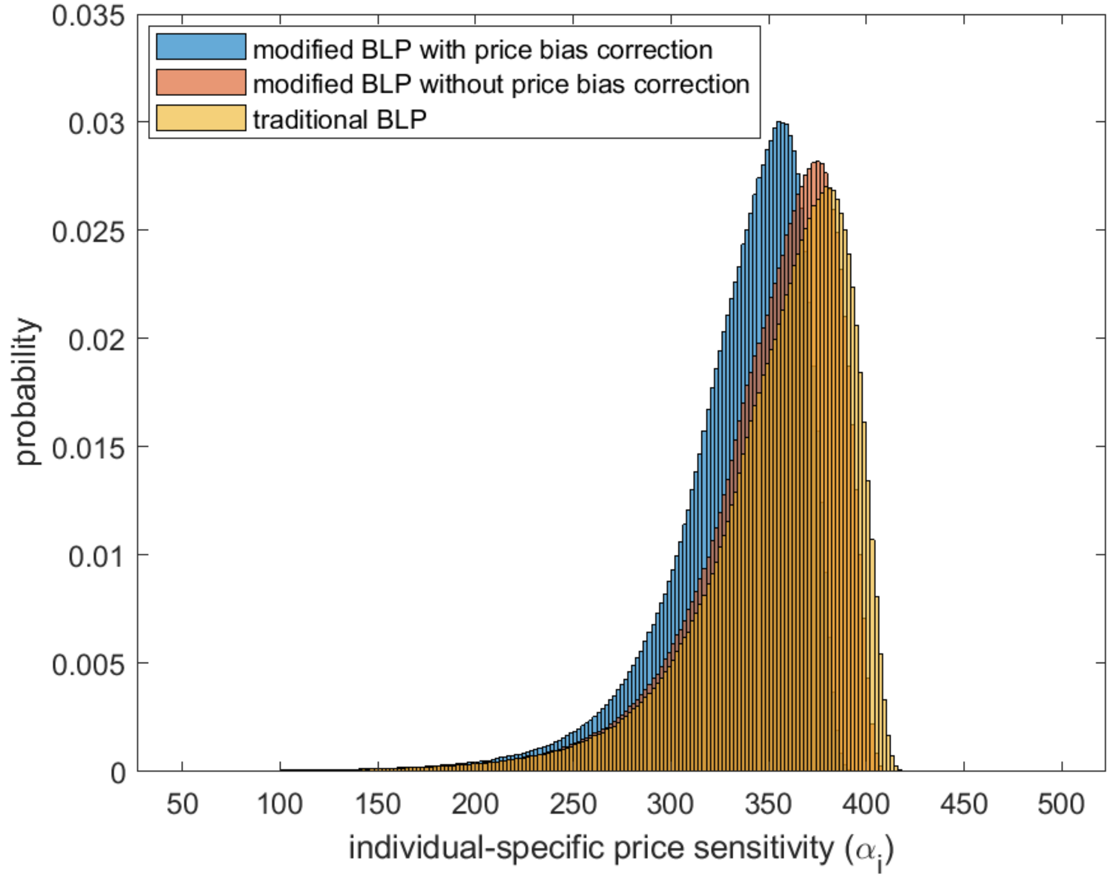

Figure 2. Distribution of individual-specific price sensitivity (α_i) (please see the color figure). The figure separately displays the distributions recovered by the modified BLP with price bias correction, the one without price bias correction, and the traditional BLP, respectively, as blue, orange, and yellow histograms. Compared with the distributions recovered by our methods, the distribution recovered by the traditional BLP method has more borrowers with high price sensitivity and fewer borrowers with low price sensitivity.

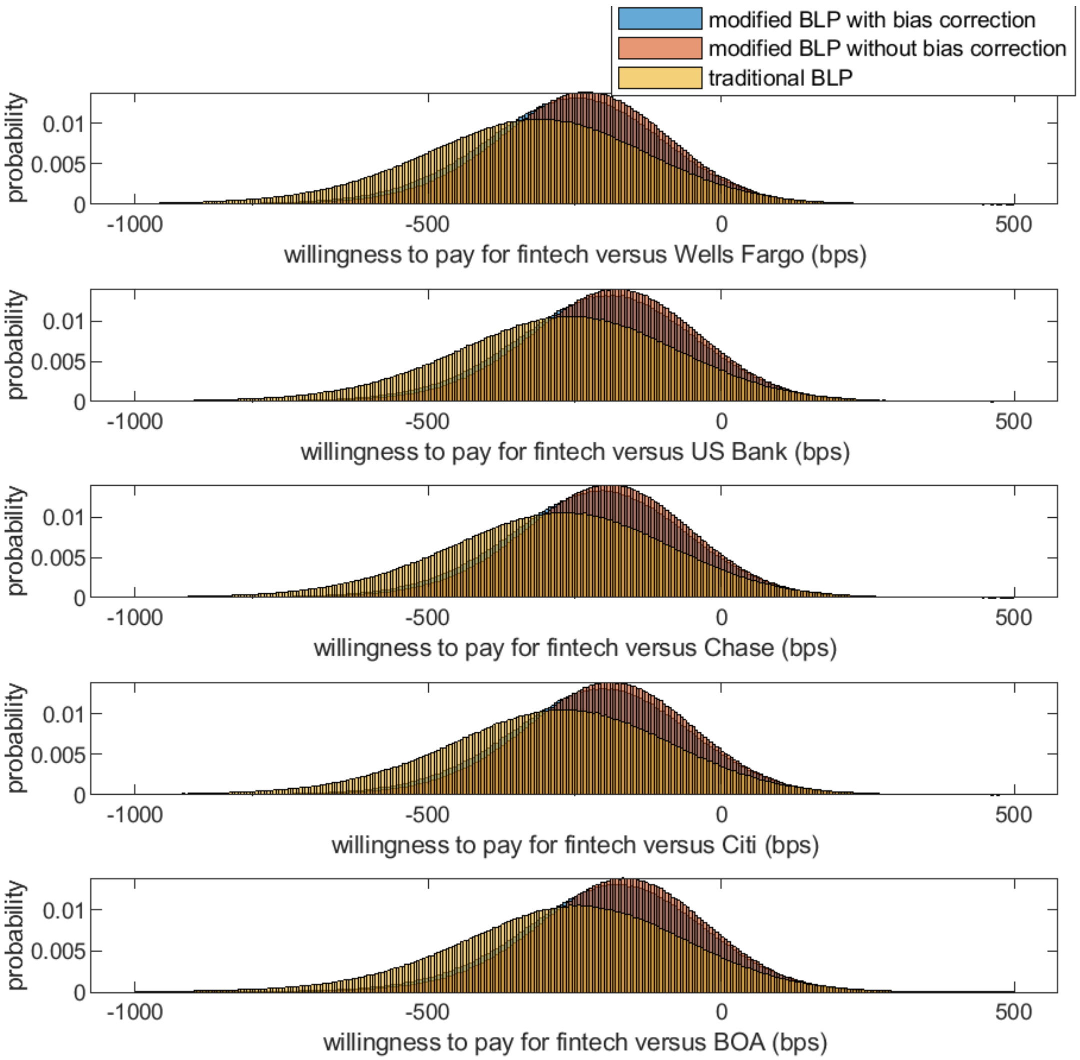

Figure 3. Distribution of borrowers' willingness to pay for choosing fintech lenders versus a certain major non-fintech lender (please see the color figure). The figure separately displays the distributions recovered by the modified BLP with price bias correction, the one without price bias correction, and the traditional BLP, respectively, as blue, orange, and yellow histograms. Compared with the distributions of willingness to pay for fintech recovered by our methods, the distribution recovered by the traditional BLP method tends to have more borrowers with low willingness to pay and fewer borrowers with high willingness to pay.

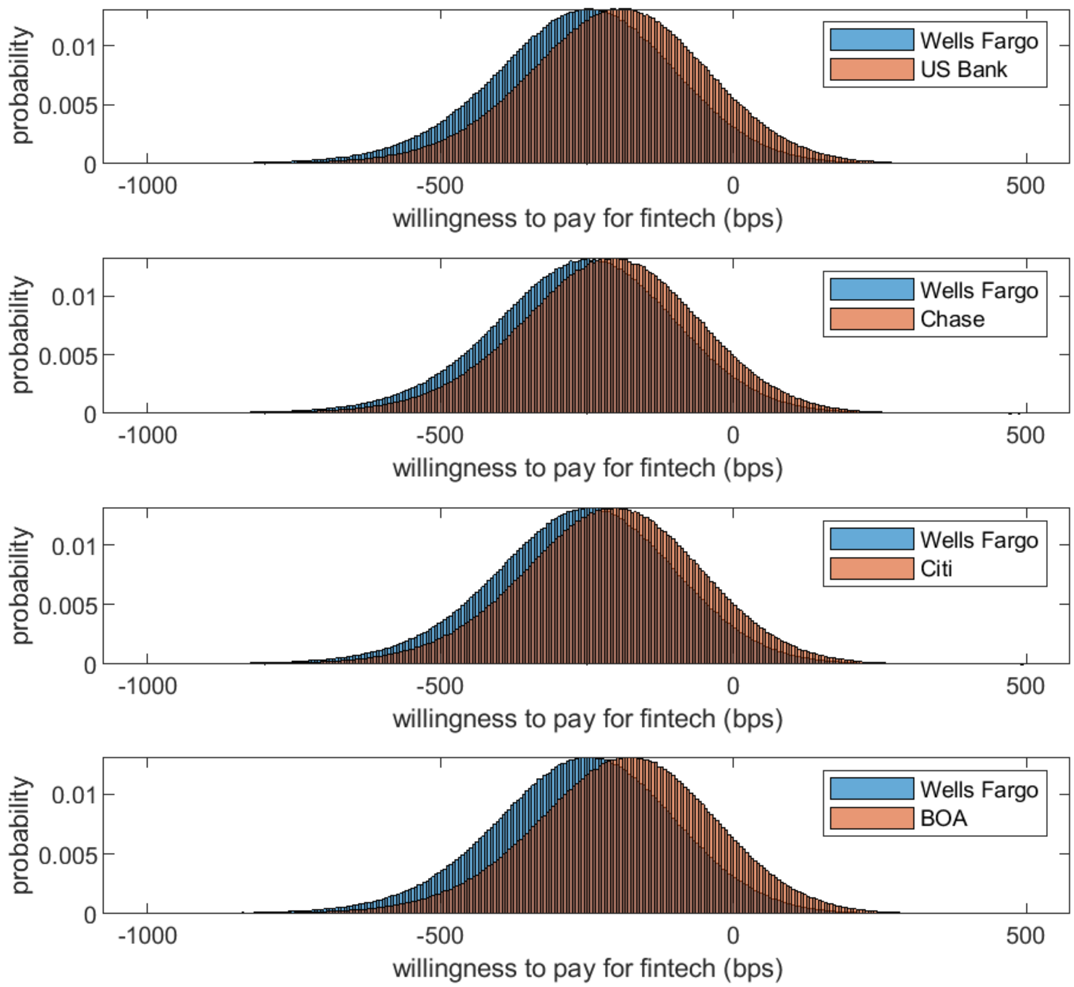

Figure 4. Comparison across major non-fintech lenders for the results generated by the modified BLP with price bias correction (please see the color figure). The distribution of borrowers' willingness to pay for choosing fintech lenders versus Wells Fargo tends to have more borrowers in the lower willingness part (the left part) than that versus any of the other four major non-fintech lenders. The reason is that Wells Fargo has the largest lender fixed effect in borrowers' utility function (see the estimate in Table 3), which is consistent with Wells Fargo's largest market share (see the descriptive statistics in Table 1).

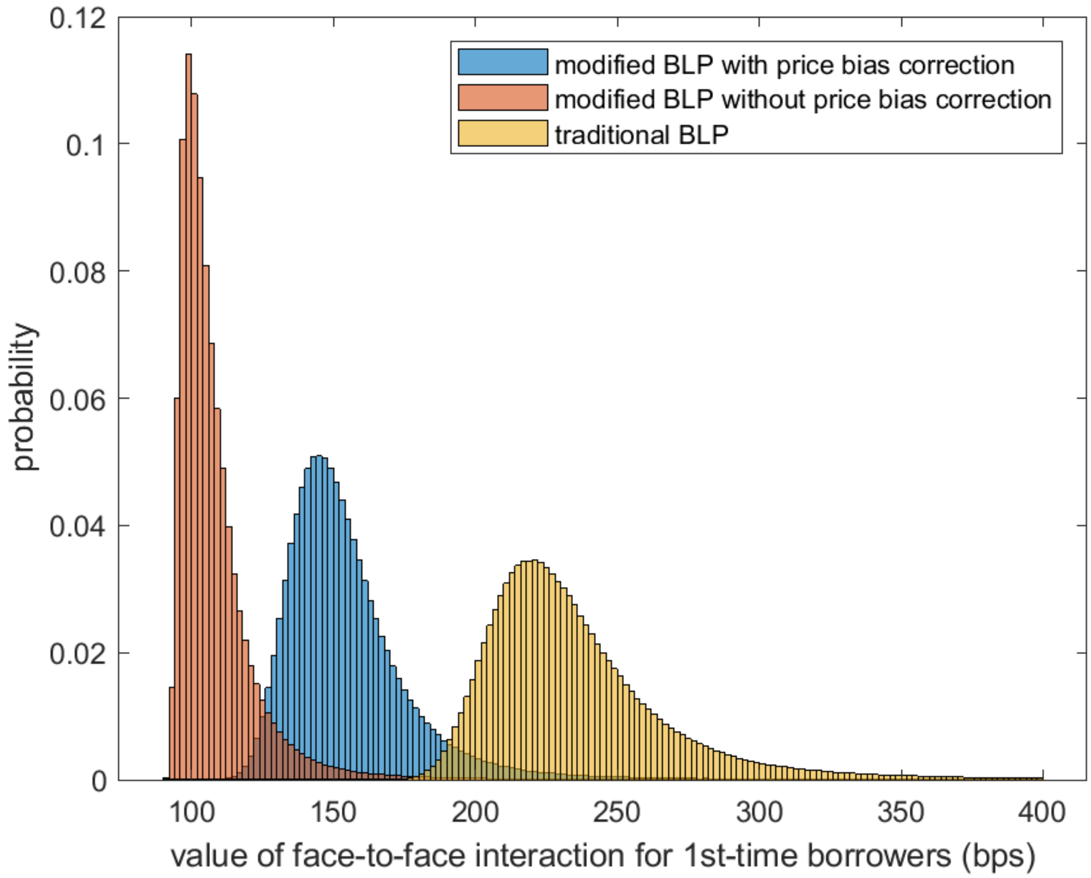

Figure 5. Distribution of the value of face-to-face interaction for first-time borrowers $\frac{\lambda_{0'}}{\alpha_i}$ (please see the color figure). The figure separately displays the distributions recovered by the modified BLP with price bias correction, the one without price bias correction, and the traditional BLP, respectively, as blue, orange, and yellow histograms.

Appendix

(For Online Publication)

Appendix A. Proofs of Theorems 1 and 2

To prove Theorems 1 and 2 in Section 4, we need to first prove Lemma 1.

Lemma 1. $\mathbb{x}_{\mathbb{j}}$ is a vector with $\mathbb{j}$ dimensions. $\mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}): R^{\mathbb{j}} \to R$ and $\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}): R^{\mathbb{j}} \to R$ are two functions. $\{\mathbb{x}_{\mathbb{j}}\}_{\mathbb{j}=1}^{+\infty}$ is a series of vectors. $\{\mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\}_{\mathbb{j}=1}^{+\infty}$ and $\{\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\}_{\mathbb{j}=1}^{+\infty}$ are two series of functions, with $0 < \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}) < +\infty$ and $0 < \mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}) < 1$ for $\forall\ \mathbb{j} \in N$ and $\forall\ \mathbb{x}_{\mathbb{j}}\ \in (-\infty, +\infty)^{\mathbb{j}}$. Meanwhile, for $\forall\ \mathbb{j} \in N$, $\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}} < +\infty$. If $\lim_{\mathbb{j}\to+\infty} \mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}) = 1$, then

$$\lim_{\mathbb{j}\to+\infty} \frac{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}}{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}} = 1$$

Proof:

Given $\lim_{\mathbb{j}\to+\infty} \mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}) = 1$, for $\forall\varepsilon > 0$, $\exists\mathbb{N} \in N$ such that for $\forall\mathbb{j} > \mathbb{N}$,

$$1 - \varepsilon < \mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}}) < 1$$

$$\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})(1-\varepsilon)\, d\mathbb{x}_{\mathbb{j}} < \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}} < \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}$$

$$(1-\varepsilon)\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}} < \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}} < \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}$$

$$1 - \varepsilon < \frac{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}}{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}} < 1$$

Therefore,

$$\lim_{\mathbb{j}\to+\infty} \frac{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\mathbb{g}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}}{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \mathbb{f}_{\mathbb{j}}(\mathbb{x}_{\mathbb{j}})\, d\mathbb{x}_{\mathbb{j}}} = 1$$

■

Theorem 1. As the number of products $J \to +\infty$,

$$\lim_{J\to+\infty} E\left[\eta_{ijmt} \middle| y_{ijmt} = 1\right] = -\alpha_i \sigma_{\eta_t}^2$$

Proof:

Using integration by parts w.r.t. η_{ijmt}, we have $\tilde{\mathfrak{H}}\left(X_j, \underline{X}_{mt}, D_i\right)$ in equation (4.2) (the numerator of $E[\eta_{ijmt}|y_{ijmt} = 1]$) as follows:

$$\tilde{\mathfrak{H}}\left(X_j, \underline{X}_{mt}, D_i\right)$$

$$= -\int \dots \int \left[\prod_{j' \neq j} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \left[\sigma_{\eta_t}^2 h\left(\frac{\eta_{ijmt}}{\sigma_{\eta_t}}\right) \frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)}\right]_{\eta_{ijmt}=-\infty}^{\eta_{ijmt}=+\infty} d\eta_{i1mt}, \dots, d\eta_{i(j-1)mt}, d\eta_{i(j+1)mt}, \dots, d\eta_{iJmt}$$

$$+ \sigma_{\eta_t}^2 \int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \left[d\left\{\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)}\right\} \Big/ d\,\eta_{ijmt}\right] d\eta_{i1mt}, \dots, d\eta_{iJmt}$$

$$= -\alpha_i \sigma_{\eta_t}^2 \int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)\left[1 + \sum_{j' \neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)\right]}{\left[1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)\right]^2} d\eta_{i1mt}, \dots, d\eta_{iJmt}$$

As the number of products $J \to +\infty$, $\dfrac{1+\sum_{j' \neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)}{1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} \to 1$.

Based on Lemma 1,

$$\frac{\tilde{\mathfrak{H}}\left(X_j, \underline{X}_{mt}, D_i\right)}{\mathfrak{H}\left(X_j, \underline{X}_{mt}, D_i\right)} \to \frac{-\alpha_i \sigma_{\eta_t}^2 \mathfrak{H}\left(X_j, \underline{X}_{mt}, D_i\right)}{\mathfrak{H}\left(X_j, \underline{X}_{mt}, D_i\right)} = -\alpha_i \sigma_{\eta_t}^2$$

Therefore,

$$\lim_{J \to +\infty} E[\eta_{ijmt}|y_{ijmt} = 1] = -\alpha_i \sigma_{\eta_t}^2$$

■

Theorem 2. As the number of products $J \to +\infty$,

$$\lim_{J \to +\infty} Var[\eta_{ijmt}|y_{ijmt} = 1] = Var[\eta_{ijmt}] = \sigma_{\eta_t}^2$$

Proof:

$$E[\eta_{ijmt}^2|y_{ijmt} = 1] = \frac{\int \dots \int \eta_{ijmt}^2 \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{iJmt}}{\int \dots \int \left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right] \frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt}, D_i\right)'\Theta_t + \eta_{ijmt}\right) + \beta X_j + \xi_{jmt}\right)}{1 + \sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt}, D_i\right)'\Theta_t + \eta_{ij'mt}\right) + \beta X_{j'} + \xi_{j'mt}\right)} d\eta_{i1mt}, \dots, d\eta_{iJmt}}$$

$$= \frac{\tilde{\tilde{\mathfrak{H}}}\left(X_j, \underline{X}_{mt}, D_i\right)}{\mathfrak{H}\left(X_j, \underline{X}_{mt}, D_i\right)}$$

where $\tilde{\tilde{\mathfrak{H}}}\left(X_j, \underline{X}_{mt}, D_i\right)$ is the numerator in $E[\eta_{ijmt}^2|y_{ijmt} = 1]$. Using integration by parts w.r.t. η_{ijmt}, we have

$$\tilde{\tilde{\mathfrak{H}}}\left(X_j,\underline{X}_{mt},D_i\right)$$

$$= -\int\dots\int \eta_{ijmt}\left[\prod_{j'\neq j} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\left[\sigma_{\eta_t}^2 h\left(\frac{\eta_{ijmt}}{\sigma_{\eta_t}}\right)\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)}{1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)}\right]_{\eta_{ijmt}=-\infty}^{\eta_{ijmt}=+\infty} d\eta_{i1mt},\dots,d\eta_{i(j-1)mt},d\eta_{i(j+1)mt},\dots,d\eta_{iJmt}$$

$$+\sigma_{\eta_t}^2\int\dots\int\left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\left[d\left\{\eta_{ijmt}\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)}{1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)}\right\}\Big/ d\,\eta_{ijmt}\right] d\eta_{i1mt},\dots,d\eta_{iJmt}$$

$$= \sigma_{\eta_t}^2\,\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)$$

$$-\alpha_i\sigma_{\eta_t}^2\int\dots\int\left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\eta_{ijmt}\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)\left[1+\sum_{j'\neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]}{\left[1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2} d\eta_{i1mt},\dots,d\eta_{iJmt}$$

$$= \sigma_{\eta_t}^2\,\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)$$

$$+\int\dots\int\left[\prod_{j'\neq j} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\left[\alpha_i\sigma_{\eta_t}^4 h\left(\frac{\eta_{ijmt}}{\sigma_{\eta_t}}\right)\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)\left[1+\sum_{j'\neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]}{\left[1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2}\right]_{\eta_{ijmt}=-\infty}^{\eta_{ijmt}=+\infty}\prod_{j'\neq j} d\eta_{ij'mt}$$

$$+\alpha_i^2\sigma_{\eta_t}^4\int\dots\int\left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)\left[1+\sum_{j'\neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2}{\left[1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^3} d\eta_{i1mt},\dots,d\eta_{iJmt}$$

$$= \sigma_{\eta_t}^2\,\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)$$

$$+\alpha_i^2\sigma_{\eta_t}^4\int\dots\int\left[\prod_{j'=1}^{J} h\left(\frac{\eta_{ij'mt}}{\sigma_{\eta_t}}\right)\right]\frac{exp\left(-\alpha_i\left(\left(\underline{X}_{jmt},D_i\right)'\Theta_t+\eta_{ijmt}\right)+\beta X_j+\xi_{jmt}\right)\left[1+\sum_{j'\neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2}{\left[1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^3} d\eta_{i1mt},\dots,d\eta_{iJmt}$$

As the number of products $J \to +\infty$, $\frac{\left[1+\sum_{j'\neq j} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2}{\left[1+\sum_{j'} exp\left(-\alpha_i\left(\left(\underline{X}_{j'mt},D_i\right)'\Theta_t+\eta_{ij'mt}\right)+\beta X_{j'}+\xi_{j'mt}\right)\right]^2} \to 1$. Based on Lemma 1,

$$\frac{\tilde{\tilde{\mathfrak{H}}}\left(X_j,\underline{X}_{mt},D_i\right)}{\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)} \to \frac{\left[\sigma_{\eta_t}^2+\alpha_i^2\sigma_{\eta_t}^4\right]\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)}{\underset{\sim}{\mathfrak{H}}\left(X_j,\underline{X}_{mt},D_i\right)} = \sigma_{\eta_t}^2+\alpha_i^2\sigma_{\eta_t}^4$$

Therefore,

$$\lim_{J\to+\infty} E\left[\eta_{ijmt}^2\middle|y_{ijmt}=1\right] = \sigma_{\eta_t}^2+\alpha_i^2\sigma_{\eta_t}^4$$

$$\lim_{J\to+\infty} Var\left[\eta_{ijmt}\middle|y_{ijmt}=1\right] = \lim_{J\to+\infty}\left\{E\left[\eta_{ijmt}^2\middle|y_{ijmt}=1\right]-E\left[\eta_{ijmt}\middle|y_{ijmt}=1\right]^2\right\} = \sigma_{\eta_t}^2$$

■

Appendix B. Monte Carlo Simulation

B.1. Data generating process

Without loss of generality, in the Monte Carlo simulation, we consider the national market in each month t as one market. The utility of borrower i for choosing product j in market (period) t is:

$$u_{ijt} = -(\alpha_0 - \gamma D_i)p_{ijt} + \beta_0 + \beta_1 X_j + \xi_{jt} + \varepsilon_{ijt} \tag{B.1}$$

D_i can be considered as borrower i's income. D_i in market t is generated from a normal distribution $\frac{0.1t}{T} + N(0,\,1)$; thus, one feature of the design is that each market is different in borrowers' average income. X_j can be considered as a product characteristic related to maturity or credit risk (e.g., LTV), which is generated from $N(0,\,1)$. ξ_{jt} is the unobserved product characteristic, which is generated from $N(0,\,0.1^2)$. The price of product j in month t faced by borrower i, p_{ijt}, is generated by the following formula:

$$\begin{aligned} p_{ijt} = {} & 10 + X_j + 0.01 \cdot \bar{X}_{-j} + 0.5 \cdot BaselineRate_t + 0.5 \cdot CreditSpread_t - D_i \\ & + \xi_{jt} + \eta_{ijt} \end{aligned} \tag{B.2}$$

In equation (B.2), $BaselineRate_t$ and $CreditSpread_t$ are generated from $N(0,\,1)$, and η_{ijt} is generated from $N(0,\,0.5^2)$. $\bar{X}_{-j}$ is the average characteristic of all the products other than product j in the market. The rationale for $\bar{X}_{-j}$ in equation (B.2) is that a firm's pricing strategy on a product can be affected by the characteristics of other products in the market. The rationale for D_i in equation (B.2) is that borrowers with higher incomes D_i pay lower prices (interest rates) because of lower default risk.

In equation (B.1), we set the structural parameters $\alpha_0 = 1$, $\gamma = 0.2$, $\beta_0 = -0.1$, and $\beta_1 = 0.1$.

We simulate borrower-specific prices based on equation (B.2). Using the simulated borrower-specific prices, for each product j, we estimate the following hedonic pricing equation:

$$p_{ijt} = a_j + b_j \cdot D_i + \psi_{jt} + \eta_{ijt} \tag{B.3}$$

ψ_{jt} is the time fixed effect in the regression for product j, which will capture all the relevant macroeconomic factors (baseline interest rate and credit spread). After we obtain the estimates for the coefficients a_j, b_j, and ψ_{jt} and the variance of the error term η_{ijt}, we predict the prices faced by each borrower for purchasing each product.

Finally, we use predicted individual-specific prices and borrower utilities to construct market shares for each product in each market based on equation (3.8). Note that unlike the observed actual price data, the simulated price data do not have the selectivity issue. Therefore, we do not need to conduct bias correction in estimation as discussed in Section 4.

B.2. Simulation results

We use $\{1, X_j, BaselineRate_t, CreditSpread_t, \bar{X}_{-j}\}$ as instrumental variables. We conduct estimations in nine scenarios that differ in the number of products in each market, the number of borrowers, and the number of markets. We also conduct estimations using the traditional BLP model. We use the mean of p_{ijt} for product j in market t as the product price in the traditional BLP model.

The simulation results are reported in Table B.1. In general, our method provides smaller estimates for γ (the effect of income on price sensitivity) than the traditional BLP model; our method also has smaller biases than the traditional BLP model on almost all the structural parameters.

Table B.1. Monte Carlo Simulation Results

# of markets	# of products in each mkt	# of borrowers in each mkt	Parameter	Modified BLP			Traditional BLP		
				Bias	S. D.	RMSE	Bias	S. D.	RMSE
250	25	100	α_0	0.0212	0.1056	0.1072	0.0094	0.1601	0.1596
			γ	**0.0056**	**0.0329**	**0.0332**	**0.0206**	**0.0481**	**0.0521**
			β_0	0.0764	0.4315	0.4361	0.7343	0.8114	1.0913
			β_1	0.0297	0.1955	0.1968	-0.3761	0.3251	0.3243
250	50	100	α_0	0.0148	0.1277	0.1280	0.0641	0.2412	0.2484
			γ	**0.0036**	**0.0381**	**0.0381**	**0.0376**	**0.0694**	**0.0787**
			β_0	0.0667	0.5650	0.5661	0.9847	1.3605	1.6739
			β_1	-0.0026	0.2800	0.2786	-0.3619	0.5195	0.5183
250	100	100	α_0	0.0005	0.0847	0.0843	0.0306	0.2838	0.284
			γ	**-0.0005**	**0.0262**	**0.0260**	**0.0324**	**0.0850**	**0.0905**
			β_0	-0.0036	0.4196	0.4175	0.7190	1.7320	1.8672
			β_1	0.0512	0.2484	0.2524	-0.5398	1.0235	1.0279
500	25	100	α_0	-0.0050	0.1271	0.1265	-0.0009	0.1754	0.1745
			γ	**-0.0028**	**0.0399**	**0.0398**	**0.0150**	**0.0555**	**0.0572**
			β_0	-0.0219	0.5395	0.5372	0.6939	0.9493	1.1720
			β_1	-0.0079	0.1941	0.1933	-0.4221	0.5734	0.5710
500	50	100	α_0	0.0077	0.0873	0.0872	0.0070	0.1760	0.1753
			γ	**0.0016**	**0.0270**	**0.0269**	**0.0208**	**0.0602**	**0.0634**
			β_0	0.0361	0.4018	0.4014	0.6661	0.9938	1.1923
			β_1	0.0468	0.2898	0.2921	-0.3755	0.6842	0.6812
500	100	100	α_0	0.0035	0.0750	0.0748	0.0263	0.2459	0.2461
			γ	**0.0007**	**0.0217**	**0.0217**	**0.0317**	**0.0799**	**0.0855**
			β_0	0.0188	0.4185	0.4168	0.6662	1.4742	1.6110
			β_1	0.0554	0.3916	0.3936	-0.4126	1.0974	1.0920
250	25	500	α_0	0.0005	0.1154	0.1149	0.0398	0.1821	0.1855
			γ	**-0.0009**	**0.0344**	**0.0343**	**0.0254**	**0.0583**	**0.0633**
			β_0	-0.0107	0.4728	0.4706	0.9197	0.9934	1.3501
			β_1	-0.0219	0.1775	0.1780	-0.3933	0.2456	0.2445
250	50	500	α_0	-0.0130	0.0921	0.0926	-0.0214	0.2278	0.2277
			γ	**-0.0046**	**0.0296**	**0.0298**	**0.0097**	**0.0756**	**0.0759**
			β_0	-0.0678	0.4280	0.4312	0.5064	1.2910	1.3808
			β_1	-0.0173	0.1733	0.1733	-0.3776	0.3678	0.3666
250	100	500	α_0	0.0005	0.0730	0.0726	0.0716	0.2373	0.2468
			γ	**-0.0003**	**0.0214**	**0.0213**	**0.0461**	**0.0675**	**0.0814**
			β_0	0.0053	0.3622	0.3605	0.9732	1.4409	1.7328
			β_1	0.0019	0.2010	0.2000	-0.4005	0.5084	0.5059

True parameter values in DGP: $\alpha_0 = 1$, $\gamma = 0.2$, $\beta_0 = -0.1$, and $\beta_1 = 0.1$. Number of Repetitions = 100.

Appendix C. Additional figures and tables

Table C.1. Quantifying the value of face-to-face interaction for first-time borrowers: the first-step estimation

Dependent variable: interest rate	Column 1 January, 2012	Column 2 June, 2012	Column 3 December, 2012
LTV			
$20\% < LTV \leq 40\%$	-0.0004	-0.0005**	-0.0005**
	(0.0003)	(0.0002)	(0.0002)
$40\% < LTV \leq 60\%$	-0.0006*	-0.0007***	-0.0008***
	(0.0003)	(0.0002)	(0.0002)
$60\% < LTV \leq 80\%$	0.0003	0.0001	0.0001
	(0.0003)	(0.0002)	(0.0002)
$LTV > 80\%$	0.0011***	0.0007***	0.0007***
	(0.0003)	(0.0002)	(0.0002)
Loan term			
15 years	-0.0037***	-0.0039***	-0.0044***
	(0.0001)	(0.0001)	(0.0001)
30 years	0.0030***	0.0036***	0.0023***
	(0.0001)	(0.0001)	(0.0001)
Lender			
Bank of America	-0.0014***	-0.0009***	-0.0008***
	(0.0002)	(0.0001)	(0.0001)
Citi	-0.0015***	-0.0014***	-0.0008***
	(0.0002)	(0.0001)	(0.0001)
Chase	-0.0006***	-0.0004***	-0.0009***
	(0.0002)	(0.0001)	(0.0001)
U.S. Bank	-0.0017***	-0.0009***	-0.0005***
	(0.0002)	(0.0001)	(0.0001)
Wells Fargo	-0.0017***	-0.0013***	-0.0012***
	(0.0002)	(0.0001)	(0.0001)
Other non-fintech lenders	-0.0010***	-0.0011***	-0.0012***
	(0.0002)	(0.0001)	(0.0001)
FICO/1000	-0.0093*	-0.0185***	-0.0201***
	(0.0055)	(0.0003)	(0.0003)
Monthly income ($ 1,000)	-0.0001***	-0.0001***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)
1st-time homebuyer	0.0002***	0.0001***	0.0001***
	(0.0001)	(0.0000)	(0.0000)
Primary residence	0.0052***	0.0045***	0.0041***
	(0.0001)	(0.0001)	(0.0001)
Investment property	-0.0006***	-0.0005***	-0.0004***
	(0.0001)	(0.0000)	(0.0000)
MSA fixed effects	✓	✓	✓
Root mean squared error $\hat{\sigma}_{\eta_t}$	0.0027	0.0023	0.0023
R squared	0.5590	0.6011	0.5730
N	39817	85590	69330

For each month t, we estimate the hedonic pricing equation (3.6). Compared with Table 2, the regressions reported in this table have additional regressors: the 1st-time homebuyer indicator and the property occupancy status categories (primary residence, second home, and investment property, with second home as the omitted reference group). Columns 1, 2, and 3 in this table report the regression results using loans originated in January, June, and December, respectively, in 2012. Regression results using loans originated in other months are available upon request. Standard errors are reported in parentheses. * denotes significance at a 10% level. ** denotes significance at a 5% level. *** denotes significance at a 1% level. Loans with $LTV \leq 20\%$, with loan terms other than 15 or 30 years, or originated by fintech lenders are the omitted reference groups.

Table C.2. Regression of hedonic pricing equation: without controlling for property occupancy status

Dependent variable: interest rate	Column 1 January, 2012	Column 2 June, 2012	Column 3 December, 2012
LTV			
$20\% < LTV \leq 40\%$	-0.0001	-0.0002	-0.0002
	(0.0004)	(0.0002)	(0.0002)
$40\% < LTV \leq 60\%$	0.0001	-0.0002	-0.0002
	(0.0004)	(0.0002)	(0.0002)
$60\% < LTV \leq 80\%$	0.0018***	0.0011***	0.0012***
	(0.0004)	(0.0002)	(0.0002)
$LTV > 80\%$	0.0015***	0.0011***	0.0011***
	(0.0004)	(0.0002)	(0.0002)
Loan term			
15 years	-0.0035***	-0.0037***	-0.0042***
	(0.0002)	(0.0001)	(0.0001)
30 years	0.0030***	0.0034***	0.0023***
	(0.0001)	(0.0001)	(0.0001)
Lender			
Bank of America	-0.0016***	-0.0010***	-0.0010***
	(0.0002)	(0.0001)	(0.0001)
Citi	-0.0017***	-0.0015***	-0.0009***
	(0.0003)	(0.0001)	(0.0001)
Chase	-0.0012***	-0.0006***	-0.0013***
	(0.0002)	(0.0001)	(0.0001)
U.S. Bank	-0.0019***	-0.0012***	-0.0007***
	(0.0003)	(0.0001)	(0.0001)
Wells Fargo	-0.0024***	-0.0017***	-0.0015***
	(0.0002)	(0.0001)	(0.0001)
Other non-fintech lenders	-0.0014***	-0.0013***	-0.0015***
	(0.0002)	(0.0001)	(0.0001)
FICO/1000	-0.0091*	-0.0175***	-0.0197***
	(0.0053)	(0.0003)	(0.0003)
Monthly income ($ 1,000)	-0.0001***	-0.0001***	-0.0001***
	(0.0000)	(0.0000)	(0.0000)
1st-time homebuyer	-0.0010***	-0.0006***	-0.0007***
	(0.0001)	(0.0000)	(0.0000)
MSA fixed effects	✓	✓	✓
Root mean squared error $\hat{\sigma}_{\eta_t}$	0.0033	0.0027	0.0027
R squared	0.3623	0.4633	0.4295
N	39817	85590	69330

This table is similar to Table C.1 but the property occupancy status is not controlled for. Standard errors are reported in parentheses. * denotes significance at a 10% level. ** denotes significance at a 5% level. *** denotes significance at a 1% level. Loans with $LTV \leq 20\%$, with loan terms other than 15 or 30 years, or originated by fintech lenders are the omitted reference groups.

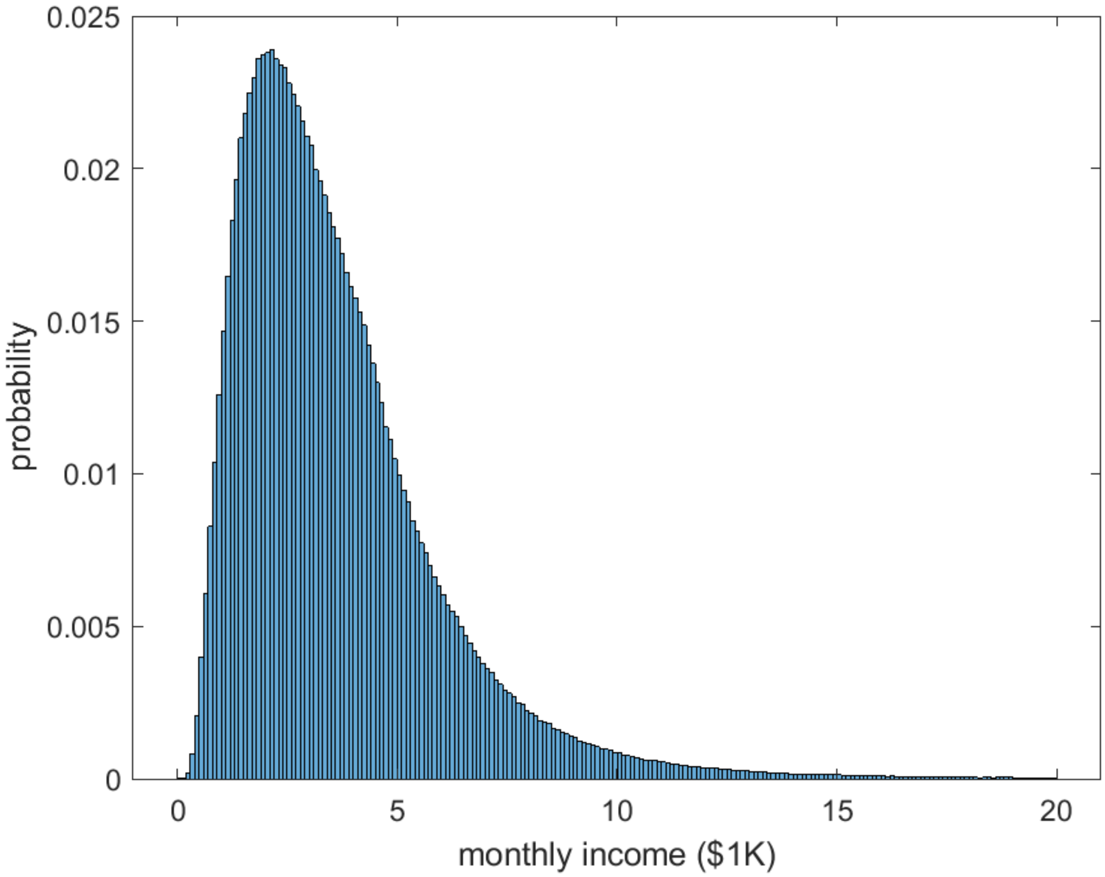

Figure C.1. Distribution of deflated monthly income (dollars in January 2010).